SIBDNet: Spectrum Information Bottleneck Decoupling Network for Domain Generalization in Cross-Scene Hyperspectral Image Classification

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***Abstract*—Due to the difficulty in timely obtaining labeled samples from the target domain in hyperspectral image (HSI), domain generalization (DG)-based methods for cross-scene HSI classification have increasingly attracted research attention. However, existing DG methods struggle to effectively disentangle class-relevant features from domain-specific ones, particularly when faced with the challenges of high-dimensional spectral–spatial coupling and domain shift. To address the above challenge, the spectrum information bottleneck decoupling network (SIBDNet) for DG is proposed. By incorporating discrete wavelet transform (DWT)-based spectrum decomposition and an information bottleneck strategy, the proposed method achieves deep disentanglement between spectral–spatial invariant features that encode class-relevant information and domain-sensitive features that capture instance-specific characteristics, thereby enhancing the generalization performance of HSI classification across diverse scenes. Specifically, the DWT is first employed to decompose HSI into low-frequency and high-frequency components (HFCs). This process extracts comprehensive physical information from the spectrum domain of HSI, capturing spectral–spatial coupled features while simultaneously mitigating interference noise from coupled spectral–spatial information. Second, the information bottleneck encoder is constructed to decouple knowledge from low-frequency and HFCs. This process provides spectral–spatial invariant features and domain-sensitive features for cross-scene knowledge transfer, whereby comprehensive extraction of source domain knowledge is achieved while mitigating spectral–spatial information coupling challenges. Finally, the Patch-Mix-based virtual domain generation method is proposed. By applying controlled perturbations to domain-sensitive features, it broadens decision boundaries to ensure diversity and effectiveness of virtual domains, thus enhancing model generalization capability toward latent unknown domains. Experimental results on multiple cross-scene HSI datasets demonstrate that the proposed method establishes state-of-the-art performance.**

***Index Terms*—Decoupled representation learning, domain generalization (DG), hyperspectral image (HSI) classification, information bottleneck, virtual domain generation.**

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I. Introduction

HYPERSPECTRAL image (HSI) captures nanoscale spectral responses across hundreds of contiguous spectral bands of surface materials [1], [2]. This unique high-dimensional spectral–spatial coupled characteristic enables its critical role in environmental protection [3], modern agriculture [4], and geological exploration [5]. As a core HSI interpretation task, HSI classification aims to establish mapping relationships from high-dimensional spectral cubes to pixel-wise land-cover labels. Its accuracy directly determines the effectiveness of HSI in practical applications, including environmental monitoring, precision agriculture, and disaster assessment [6].

In recent years, deep learning-based HSI classification methods have significantly enhanced accuracy through joint spectral–spatial feature learning [7], [8]. However, such high precision critically relies on the assumption that training and test datasets satisfy independent and identically distributed conditions. In practical HSI classification scenes, due to the high costs of HSI data acquisition and annotation, researchers typically aim to classify unlabeled target domains using models trained on source domains with abundant labeled samples. Within such cross-scene HSI classification tasks, deep learning methods that excel in source domains exhibit poor generalization to target domains, with their accuracy constrained by domain shift (DS) challenges arising from spectral drift and spatial distribution disparities [9]. Domain adaptation (DA) methods mitigate classification performance degradation caused by DS through explicit feature distribution alignment between source and target domains during training [10]. Nevertheless, their dependency on target domain data during training fails to meet practical application requirements, for instance, disaster response scenarios (e.g., wildfires, floods) demand models capable of processing geographically unknown regions during training [11].

Given the inherent limitation of DA methods in requiring target domain data during training, the recent research paradigm for addressing DS challenges is progressively shifting toward domain generalization (DG) methods [12]. DG constructs domain-invariant models solely using labeled source

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domain data, enabling cross-scene inference without necessitating target domain prior knowledge during training. This delivers more practical solutions for HSI classification in real-world applications. Current mainstream DG methods include data augmentation-based methods and domain-invariant learning methods. Data augmentation-based methods generate virtual domain samples exhibiting DS characteristics through perturbation strategies such as spectral–spatial randomization and structural deformation. This mitigates model overfitting to limited source domain data distributions while ensuring sufficient coverage of potential unseen target domains in feature space [13], [14]. Domain-invariant learning methods establish robust feature spaces resistant to DS via physics-driven constraints or semantic alignment mechanisms, facilitating cross-scene transfer of domain-invariant features to alleviate DS challenges when confronting unknown target domains [15], [16].

However, these methods still face challenges in addressing cross-scene coupled shift features in HSI due to the absence of decoupling mechanisms for class-relevant domain-invariant features and sample-specific domain-sensitive features. Without effective feature decoupling, data augmentation-based DG methods inevitably compromise class-relevant domain-invariant features when perturbing domain-specific characteristics of samples, leading to diminished class separability. Therefore, deep decoupling of complex spectral–spatial coupled features in HSI is proposed. The decoupling process isolates domain-specific features from samples, upon which perturbation strategies are applied to enhance the diversity and sufficiency of virtual domains generated for DG tasks. Furthermore, given the general lack of physics-driven constraints with strong priors in HSI datasets, domain-invariant learning methods encounter limitations: without implementing spectral–spatial feature decoupling, domain-specific characteristics inevitably interfere with constructing cross-scene invariant feature spaces, resulting in suboptimal performance. Consequently, leveraging decoupled class-relevant domain-invariant features, this work constructs cross-scene invariant feature spaces to improve the authenticity and effectiveness of generated virtual domains, thereby enhancing model capability for cross-scene knowledge transfer.

Concretely, the spectrum information bottleneck decoupling network (SIBDNet) for cross-scene HSI classification is proposed. First, the frequency-domain physics-informed discrete wavelet transform (DWT) spectrum decomposition method parses HSI into low-frequency components (LFCs) and high-frequency components (HFCs). LFC preserves essential attributes of land-cover reflectance spectra through low-pass filtering, while HFC captures structured spectral–spatial coupled information via high-pass filtering. Subsequently, the information bottleneck encoder (IBE) decouples knowledge from LFC and HFC to obtain spectral–spatial invariant features and domain-sensitive features. Finally, the Patch-Mix-based virtual domain generation method applies controlled perturbations to domain-sensitive features to generate virtual samples. The diversity and representational efficacy of virtual domains are ensured through supervised contrastive learning and adversarial mechanisms. By approximating potential unseen target domains with generated virtual domains, comprehensive coverage of target domain feature distributions is achieved, thereby mitigating DS challenges and enhancing model generalization capability for latent unknown target domains. The contributions are summarized below.

1) The SIBDNet is proposed. By integrating information bottleneck knowledge decoupling and virtual domain expansion, this simultaneously addresses the challenges of modeling complex spectral–spatial coupled features and transferring knowledge under DS.
2) The DWT frequency spectrum decomposition method is developed. This separates frequency spectrum dimensional information by decoupling structured high-frequency spectral–spatial coupled components from low-frequency land-cover reflectance spectral attributes. It maintains spectral–spatial feature modeling capability while enabling effective extraction of spectrum components.
3) The information bottleneck knowledge decoupling architecture is designed. Building upon spectrum components, this achieves an upgrade from shallow physical decoupling to deep information decoupling. It provides spectral–spatial invariant features alongside domain-sensitive features for cross-scene knowledge transfer.
4) The Patch-Mix-based virtual domain generation method is developed. By applying controlled perturbations to domain-sensitive features, this expands the coverage of decision boundaries in feature space. It enhances the adaptability for cross-scene knowledge transfer to target domains.

II. Related Work

A. DA for Cross-Scene HSI Classification

Given the high cost of pixel-level HSI annotation, target domains often possess few or even no labeled samples, which is insufficient to support deep model training [17], [18], [19]. To overcome this limitation, researchers have introduced DA techniques that enable cross-scene knowledge transfer and enhance model generalization by measuring and narrowing distribution gaps between source and target domains [20], [21], [22].

Ding et al. [23] proposed a diffusion-augmented prototype knowledge distillation DA method. By integrating diffusion-enhanced unsupervised pretraining, dual-branch spectral–spatial attention mechanisms, and global–local prototype knowledge distillation, this method effectively mitigates DS and noise interference under extreme scarcity of target domain annotations. Feng et al. [24] introduced a class-aligned and class-balancing generative DA method. It addresses class imbalance and cross-domain misalignment in HSI classification through class-aligned adversarial loss, class-balanced data sampling, and spectral–spatial equivariant feature extraction via split-level CapsNet. Luo et al. [25] developed a prototype-guided DA method. This achieves high-precision cross-scene HSI classification with minimal annotation budgets by initial prototype alignment, inconsistency-aware sample selection, and class-balanced self-training. Ding et al. [26] presented a consistency-aware customized learning method. It enables efficient cross-scene knowledge transfer through adversarial

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domain alignment, prototype-based sample differentiation, and customized cross/intradomain contrastive learning.

Current DA methods for cross-scene HSI classification predominantly require customized model architectures and training procedures for each target scene, mandating access to target domain data during training. However, this assumption significantly deviates from real-world remote sensing operational senses. In polar regions, foreign territories, sensitive areas, or emergency zones, target domain imagery is often unavailable for real-time acquisition or cannot be transmitted due to security restrictions. Simultaneously, repeated data collection and model retraining for each new region incur prohibitive temporal, computational, and labor costs, posing substantial deployment challenges for DA technologies in practical applications.

B. DG for Cross-Scene HSI Classification

DG techniques are proposed to address the practical challenge of numerous target scenes with inaccessible target domain data. Current mainstream DG methods follow two primary pathways: 1) data augmentation-based methods that generate or extrapolate source distributions to cover potential target patterns and 2) domain-invariant learning methods that directly extract features robust to domain variations for cross-scene HSI classification.

Data augmentation-based DG methods artificially expand source domain distributions through diversified perturbations, mixing, or generation techniques. This extends feature space boundaries to encompass potential target domain patterns, significantly enhancing model generalization to unseen scenes without target domain samples [27], [28]. Zhang et al. [13] proposed a single-source domain expansion network (SDEnet), where dual-branch generators generate expanded domain samples with both DS and discriminative properties. Combined with supervised contrastive adversarial learning, this enables robust cross-scene HSI classification using only source data. Zhao et al. [29] introduced a locally linear unbiased randomization network (LLURnet), generating expanded samples via symmetric encoder-randomization-decoder structures for nonredundant, class-separable augmentation. Qing et al. [14] developed a frequency disentanglement and data geometry network (FDGNet), preserving high-frequency semantics while reconstructing low-frequency textures with manifold geometry constraints for target-free cross-scene classification. These methods face a critical limitation: the absence of explicit decoupling between class-relevant domain-invariant features and sample-specific domain-sensitive features. When confronting coupled spectral–spatial shifts in HSI, indiscriminate perturbation of domain-specific components compromises class-relevant invariant structures, degrading class separability and ultimately undermining generalization performance.

Domain-invariant learning-based DG methods focus on extracting domain-invariant features from source domains, constructing robust cross-scene shareable representations to enable unbiased knowledge transfer to arbitrary unseen target domains [30]. Zhang et al. [15] proposed a language-aware DG network (LDGnet) that aligns visual–textual modalities through dual encoders and supervised contrastive learning, leveraging linguistic space as a cross-domain semantic bridge for class-discriminative invariant representations. Dang et al. [31] developed a linguistic-interact-with-visual engager network, which extracts global context and class correlations, achieving deep cross-modal alignment via its attention modules for enhanced discriminative transferability. Wang et al. [16] introduced a text-driven adaptive semantic alignment network that maps visual features to a shared semantic space constructed by template prompts and vision-guided fine-grained prompts. Despite constructing cross-domain robust feature spaces through textual modalities, these methods fail to decouple spectral–spatial features. Domain-specific features remain intertwined with invariant representations, compromising feature space efficacy and ultimately undermining transfer capability.

This work leverages DWT to initially decompose HSI in the frequency domain into LFC, preserving essential land-cover spectral attributes and HFC characterizing spectral–spatial structural coupling. This avoids coupling interference during direct feature extraction. The IBE then performs deep decoupling on spectrum components, yielding spectral–spatial invariant features and domain-sensitive features. The spectral–spatial invariant features model class-discriminative information and maintain cross-scene stability. When employed in domain-invariant learning, they remain unaffected by domain-specific features, ensuring virtual domain validity and consequently enhancing generalization in cross-scene knowledge transfer. Domain-sensitive features fully encapsulate source-domain-specific attributes. Applying targeted perturbations to these features enables more effective expansion of the feature space, increases virtual domain diversity, and achieves comprehensive coverage of potential target domain distributions, ultimately strengthening robustness in cross-scene knowledge transfer.

C. Disentanglement-Based DG Methods

Research on disentanglement-based DG has seen significant progress in the field of computer vision (CV). Fan et al. [32] introduced a DG method based on adaptive normalization, where normalization statistics are learned by a neural network to decouple feature representations from domain-specific statistical properties, thus improving cross-DG under single-source domain training. Zhang et al. [33] proposed a DG method based on feature distribution disentanglement, which achieves the decoupling of feature representations from specific domain styles by precisely matching the empirical cumulative distribution functions of features, thereby enhancing model generalization on unseen domains. Wan et al. [34] presented a DG method based on meta-feature decomposition, which disentangles feature representations from specific domain information by representing convolutional features as a combination of universal meta-features, enhancing cross-DG for models trained on a single source domain.

However, HSI presents unique challenges: its content is highly spectral-spatially coupled, making it difficult to cleanly separate semantic content from visual style as in natural images. Directly applying CV disentanglement strategies may yield limited effectiveness. SIBDNet addresses this by employing 3-D DWT spectral decomposition, which provides

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physical prior guidance for disentanglement by preliminarily separating the coupled spectral–spatial information into low-frequency and HFCs. Subsequently, deep purification and disentanglement at the information level are achieved through the information bottleneck. This method, combining physics-guided and information-theory-driven disentanglement, represents an advancement of CV disentanglement concepts tailored to the features of HSI data.

D. Theoretical Foundations of the Information Bottleneck and Its Deep Learning Applications

The information bottleneck method proposed by Tishby et al. [35] provides an information-theoretic foundation for understanding representation learning in deep learning. Its core idea is to learn a compressed representation of the input $\boldsymbol{X}$ that maximizes the preservation of information relevant to the output $\boldsymbol{Y}$ while minimizing other information. Alemi et al. [36] introduced the deep variational information bottleneck, which combines the information bottleneck principle with variational inference to derive an objective function readily optimized by deep neural networks. Wang et al. [37] shifted the target of information compression from feature representations to model weights, establishing a weight information bottleneck based on PAC-Bayes generalization theory, thereby more directly linking it to model generalization capability. Cheng et al. [38] proposed a contrastive log-ratio upper bound method for mutual information estimation, providing a more stable and efficient tool for estimating and optimizing mutual information minimization problems. Xu et al. [39] proposed an adversarial training method based on the information bottleneck method, termed InfoAT. It systematically enhances the model's adversarial robustness by identifying and focusing on strengthening those vulnerable samples while preserving robust features within adversarial examples.

Within this context, SIBDNet applies the information bottleneck principle to the feature disentanglement problem in cross-scene HSI classification. We not only leverage its capabilities for compression and information preservation but, more importantly, extend its objective function. By introducing a constraint term specifically targeting the domain-sensitive features $\boldsymbol{H}$, we explicitly enforce orthogonal disentanglement between the spectral–spatial invariant features and the domain-sensitive features. This provides a novel application paradigm for the information bottleneck in addressing DG problems, particularly for handling complex, coupled data such as HSI.

III. Method

A. Framework of SIBDNet

Let the source domain HSI dataset be denoted as $\mathbb{S} = \{\boldsymbol{x}_i^s, \boldsymbol{y}_i\}_{i=1}^{N_s}$ and the target domain dataset as $\mathbb{T} = \{\boldsymbol{x}_i^t, \boldsymbol{y}_i\}_{i=1}^{N_t}$, $\boldsymbol{x}_i^s$ and $\boldsymbol{x}_i^t$ represent the source domain and target domain data, respectively, and $\boldsymbol{y}_i$ denotes class labels. As illustrated in Fig. 1, the training pipeline comprises of SIBDNet:

1) DWT-Based Knowledge Decoupling: First, source domain data is processed by the 3-D DWT decoupling subnet. Through high-pass and low-pass filtering decomposition, this generates a spectrum decoupling pair $\langle \boldsymbol{H}, \boldsymbol{L} \rangle$ comprising HFC and LFC. The $\langle \boldsymbol{H}, \boldsymbol{L} \rangle$ is then fed into the IBE, which optimizes feature representations by extracting critical discriminative information while suppressing redundant elements. Finally, spectral–spatial invariant features and domain-sensitive features are sampled from the refined feature space.

2) Patch-Mix-Based Virtual Domain Generation: First, spectral–spatial invariant features are processed through the spectral–spatial joint encoder, yielding spectral–spatial joint features $\boldsymbol{Z}_{\text{spe-spa}}$. Domain-sensitive features then undergo Patch-Mix perturbation to generate mixed features $\boldsymbol{Z}_{\text{Mix}}$. Finally, the integrated spectral–spatial features and mixed features are randomly fused to generate virtual domains.

3) Adversarial Optimization Learning: First, source and virtual domain samples are fed into the discriminator, outputting classified probability distributions and deep discriminative features. The discriminator parameters are then optimized using classification loss and supervised contrastive loss to enhance discriminability, while the generator is updated via adversarial loss and generative similarity loss to improve synthesis quality. Finally, optimal model parameters are obtained through iterative adversarial optimization between the discriminator and generator.

B. 3-D DWT-Based Spectrum Decomposition

Given that HSIs typically comprise 3-D data structures with tens to hundreds of spectral bands, directly modeling feature distributions incurs high computational costs. Most prior methods employ CNNs to capture semantic features, yet the absence of predownsampling noise filtering may propagate high-frequency noise into output feature maps, degrading classification accuracy. DWT decomposes complex 3-D data into: LFC preserves fundamental reflectance attributes. HFC encodes geometric-spectral details. This frequency-aware decomposition retains critical high-frequency patterns during downsampling, enriches feature representation for capturing subtle variations, and enhances model robustness against complex scene variations through spectral–spatial decoupling.

First, DWT performs downsampling on input HSI $\boldsymbol{X} \in \mathbb{R}^{H\times W\times B}$. The operation utilizes Haar wavelets with low-pass filter Φ_L and high-pass filter Φ_H, defined as

$$\begin{cases} \Phi_L = \dfrac{1}{\sqrt{2}}[1,1]^{\mathrm{T}} \\ \Phi_H = \dfrac{1}{\sqrt{2}}[1,-1]^{\mathrm{T}}. \end{cases} \tag{1}$$

The decomposition yields four subbands, formally expressed as

$$\boldsymbol{X}_{\mathrm{LL}}, \{\boldsymbol{X}_{\mathrm{LH}}, \boldsymbol{X}_{\mathrm{HL}}, \boldsymbol{X}_{\mathrm{HH}}\} = \mathrm{DWT}(\boldsymbol{X}) \tag{2}$$

where $\boldsymbol{X}_{\mathrm{LL}}$ denotes the LFC, while $\{\boldsymbol{X}_{\mathrm{LH}}, \boldsymbol{X}_{\mathrm{HL}}, \boldsymbol{X}_{\mathrm{HH}}\}$ represent HFC in vertical, horizontal and diagonal orientations.

To address the 3-D structural properties of HSI, we extend DWT to 3-D implementation by performing wavelet decomposition along both spatial and spectral dimensions. This 3-D DWT refinement significantly enhances spectral–spatial feature representation capabilities. The resulting spatial subbands $X_{\mathrm{LL}}^{\mathrm{spa}}, \{\boldsymbol{X}_{\mathrm{LH}}^{\mathrm{spa}}, \boldsymbol{X}_{\mathrm{HL}}^{\mathrm{spa}}, \boldsymbol{X}_{\mathrm{HH}}^{\mathrm{spa}}\}$ and spectral subbands $\boldsymbol{X}_{\mathrm{LL}}^{\mathrm{spe}}, \{\boldsymbol{X}_{\mathrm{LH}}^{\mathrm{spe}}, \boldsymbol{X}_{\mathrm{HL}}^{\mathrm{spe}}, \boldsymbol{X}_{\mathrm{HH}}^{\mathrm{spe}}\}$ are then concatenated through

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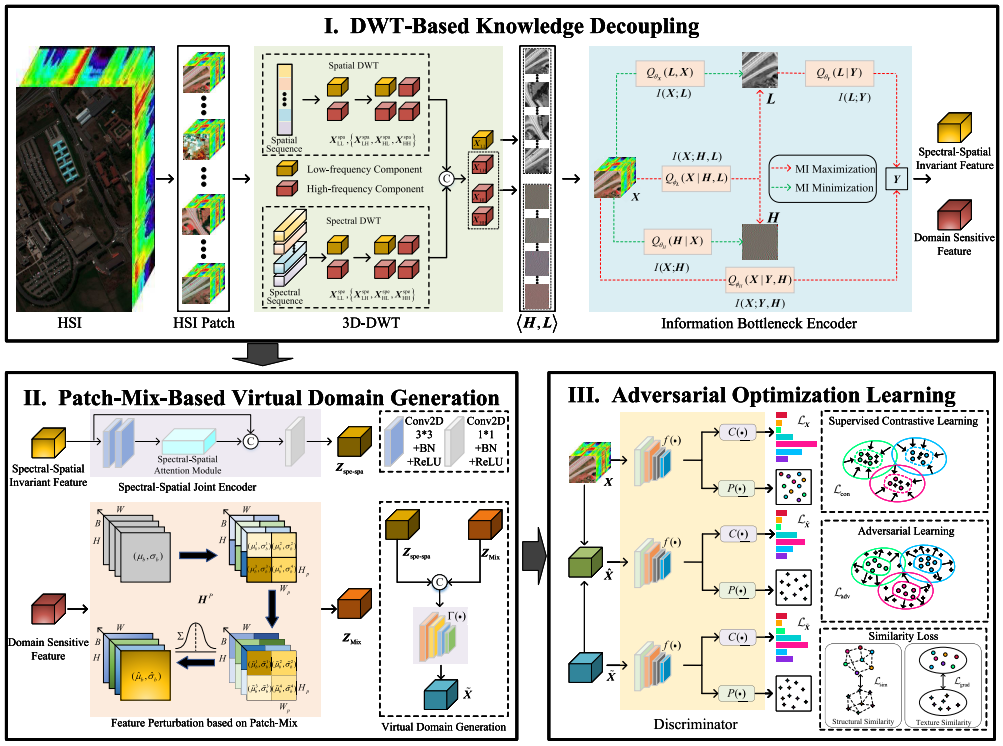


Fig. 1. Framework of SIBDNet.

channel-wise fusion, yielding the final spectrum decomposition components $\boldsymbol{X}_{\mathrm{LL}}, \{\boldsymbol{X}_{\mathrm{LH}}, \boldsymbol{X}_{\mathrm{HL}}, \boldsymbol{X}_{\mathrm{HH}}\}$.

Finally, the subbands undergo convolutional aggregation to form the spectrum decoupling pair $\langle \boldsymbol{H}, \boldsymbol{L} \rangle$

$$\begin{cases} \boldsymbol{H} = \mathrm{ReLU}\left(\mathrm{Conv}\left(\mathrm{CAT}\left(\boldsymbol{X}_{\mathrm{LH}}, \boldsymbol{X}_{\mathrm{HL}}, \boldsymbol{X}_{\mathrm{HH}}\right)\right)\right) \\ \boldsymbol{L} = \mathrm{ReLU}\left(\mathrm{Conv}\left(\boldsymbol{X}_{\mathrm{LL}}\right)\right) \end{cases} \tag{3}$$

where rectified linear unit (ReLU) $(\cdot)$ denotes the ReLU activation function, Conv$(\cdot)$ represents the convolutional operation, and CAT$(\cdot)$ indicates feature concatenation. The HFC encapsulates detailed structural information across spatial orientations, while the LFC preserves essential spectral content of HSI.

C. IBE-Based Information Bottleneck Knowledge Decoupling

To address feature decoupling in HSI classification, this method constructs a frequency-decoupled representation $\langle \boldsymbol{H}, \boldsymbol{L} \rangle$ for input HSI data $\boldsymbol{X}$ and labels $\boldsymbol{Y}$, where $\boldsymbol{L}$ encodes class-relevant spectral–spatial invariant features and $\boldsymbol{H}$ carries sample-specific domain-sensitive features. $\langle \boldsymbol{H}, \boldsymbol{L} \rangle$ under the constraint of orthogonal complementarity. Three core challenges must be resolved.

1) *Challenge 1 (Representational Completeness):* How to ensure $\boldsymbol{L}$ comprehensively models class-relevant spectral–spatial invariant attributes in $\boldsymbol{X}$.
2) *Challenge 2 (Feature Separability):* How to guide $\boldsymbol{H}$ to precisely capture sample-specific perturbations in $\boldsymbol{X}$.
3) *Challenge 3 (Orthogonal Disentanglement):* How to enforce strict decoupling between $\boldsymbol{H}$ and $\boldsymbol{L}$ to prevent feature contamination from information leakage.

To address these challenges from an information-theoretic perspective [36], we construct an information decoupling bottleneck within a unified encoder framework, solving the following optimization problem:

$$\begin{aligned} &\max_{\boldsymbol{H},\boldsymbol{L}} \mathcal{L}_{\mathrm{IDB}} \\ &\quad = \underbrace{I\left(\boldsymbol{X};\boldsymbol{H},\boldsymbol{L}\right)}_{\text{Decomposition}} + \underbrace{I\left(\boldsymbol{L};\boldsymbol{Y}\right) - \alpha I\left(\boldsymbol{X};\boldsymbol{L}\right)}_{\text{Compression}} - \underbrace{\beta I\left(\boldsymbol{Y};\boldsymbol{H}\right)}_{\text{Decoupling}} \end{aligned} \tag{4}$$

where α and β are hyperparameters controlling the compression strength and decoupling degree, respectively.

The decomposition term maximizes the mutual information $I(\boldsymbol{X};\boldsymbol{H},\boldsymbol{L})$, promoting preservation of complete $\boldsymbol{X}$ features in the decoupled representation $\langle \boldsymbol{H}, \boldsymbol{L} \rangle$ to ensure informational completeness during disentanglement.

The compression term adheres to the information bottleneck principle: By maximizing $I(\boldsymbol{L};\boldsymbol{Y})$, it forces $\boldsymbol{L}$ to encode discriminative features strongly relevant to labels $\boldsymbol{Y}$; concurrently, minimizing $I(\boldsymbol{X};\boldsymbol{L})$ suppresses redundant classification-agnostic information in $\boldsymbol{L}$. This achieves maximal compression of $\boldsymbol{X}$ by $\boldsymbol{L}$ while preserving maximal information relevant to $\boldsymbol{Y}$.

The decoupling term minimizes mutual information $I(\boldsymbol{Y};\boldsymbol{H})$ to eliminate label-related information from $\boldsymbol{H}$, confining all task-agnostic domain-sensitive features within $\boldsymbol{H}$, thereby achieving strict decoupling between components $\boldsymbol{H}$ and $\boldsymbol{L}$.

Calculating mutual information between variables is challenging when constructing the aforementioned information decoupling bottleneck; therefore, this work maximizes the optimization problem by deriving upper or lower bounds for each term.

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1) Decomposition Term: The mutual information $I(\boldsymbol{X};\boldsymbol{H},\boldsymbol{L})$ for the decomposition term is defined as

$$I(\boldsymbol{X};\boldsymbol{H},\boldsymbol{L}) = H(\boldsymbol{X}) - H(\boldsymbol{X} \mid \boldsymbol{H},\boldsymbol{L}) \tag{5}$$

where $H(\boldsymbol{X})$ is the entropy of $\boldsymbol{X}$, and $H(\boldsymbol{X}|\boldsymbol{H},\boldsymbol{L}) = -\mathbb{E}_{P(\boldsymbol{X},\boldsymbol{H},\boldsymbol{L})}[\log P(\boldsymbol{X}|\boldsymbol{H},\boldsymbol{L})]$ is the conditional entropy.

According to the idea of variational inference, the introduction of $Q_{\phi_L}(\boldsymbol{X}|\boldsymbol{H},\boldsymbol{L})$ is a variational approximation with parameters $\phi_{\boldsymbol{L}}$ to the true posterior distribution $P(\boldsymbol{X}|\boldsymbol{H},\boldsymbol{L})$. Since the kullback-leibler (KL) divergence is non-negative, i.e.,

$$D_{\mathrm{KL}}\left(P(\boldsymbol{X} \mid \boldsymbol{H},\boldsymbol{L}) \,\|\, Q_{\phi_L}(\boldsymbol{X} \mid \boldsymbol{H},\boldsymbol{L})\right) \geq 0. \tag{6}$$

After expansion, we obtain

$$\begin{aligned}&\mathbb{E}_{P(\boldsymbol{X}|\boldsymbol{H},\boldsymbol{L})}\left[\log P(\boldsymbol{X} \mid \boldsymbol{H},\boldsymbol{L})\right] \\ &\quad \geq \mathbb{E}_{P(\boldsymbol{X}|\boldsymbol{H},\boldsymbol{L})}\left[\log Q_{\phi_L}(\boldsymbol{X} \mid \boldsymbol{H},\boldsymbol{L})\right].\end{aligned} \tag{7}$$

Then the computable lower bound of $I(\boldsymbol{X};\boldsymbol{H},\boldsymbol{L})$ is expressed as

$$\begin{aligned}I(\boldsymbol{X};\boldsymbol{H},\boldsymbol{L}) &= H(\boldsymbol{X}) - H(\boldsymbol{X} \mid \boldsymbol{H},\boldsymbol{L}) \\ &\geq \mathbb{E}_{P(\boldsymbol{X},\boldsymbol{H},\boldsymbol{L})}\left[\log Q_{\phi_L}(\boldsymbol{X} \mid \boldsymbol{H},\boldsymbol{L})\right] + H(\boldsymbol{X})\end{aligned} \tag{8}$$

where $H(\boldsymbol{X})$ relates only to the data distribution $P(\boldsymbol{X})$ and requires no modeling during optimization, thus constituting a negligible term.

2) Compression Term: For the compression term, referencing the derivation process of the decomposition term, by introducing $Q_{\theta_Y}(\boldsymbol{L}|\boldsymbol{Y})$ as a variational approximation to the true posterior distribution $P(\boldsymbol{Y}|\boldsymbol{L})$, the computable lower bound of the mutual information $I(\boldsymbol{L};\boldsymbol{Y})$ is expressed as

$$\begin{aligned}I(\boldsymbol{L};\boldsymbol{Y}) &= H(\boldsymbol{L}) - H(\boldsymbol{L} \mid \boldsymbol{Y}) \\ &\geq \mathbb{E}_{P(\boldsymbol{L},\boldsymbol{Y})}\left[\log Q_{\theta_Y}(\boldsymbol{L} \mid \boldsymbol{Y})\right] + H(\boldsymbol{L})\end{aligned} \tag{9}$$

where $P(\boldsymbol{L},\boldsymbol{Y}) = P(\boldsymbol{Y},\boldsymbol{L}) = P(\boldsymbol{L}) \cdot P(\boldsymbol{Y}|\boldsymbol{L})$, and $H(\boldsymbol{L})$ is negligible.

For the mutual information $I(\boldsymbol{X};\boldsymbol{L})$, since the prior distribution of $\boldsymbol{L}$ is unknown, $r(l)$ cannot be used as a variational approximation to $P(\boldsymbol{L})$. According to [38], the variational contrastive log-upper bound (vCLUB) $I_{\mathrm{vCLUB}}(\boldsymbol{X};\boldsymbol{L})$ is employed as a variational approximation for $I(\boldsymbol{X};\boldsymbol{L})$, where

$$\begin{aligned}I_{\mathrm{vCLUB}}(\boldsymbol{X};\boldsymbol{L}) &= \mathbb{E}_{P(\boldsymbol{X},\boldsymbol{L})}\left[\log Q_{\theta_X}(\boldsymbol{L}|\boldsymbol{X})\right] \\ &\quad - \mathbb{E}_{P(\boldsymbol{X})}\mathbb{E}_{P(\boldsymbol{L})}\left[\log Q_{\theta_X}(\boldsymbol{L}|\boldsymbol{X})\right]\end{aligned} \tag{10}$$

where $Q_{\theta_X}(\boldsymbol{L}|\boldsymbol{X})$ is a variational approximation to the true posterior distribution $P(\boldsymbol{L}|\boldsymbol{X})$.

a) Upper bound derivation: For the difference between $I_{\mathrm{vCLUB}}(\boldsymbol{X};\boldsymbol{L})$ and $I(\boldsymbol{X};\boldsymbol{L})$

$$\begin{aligned}\Delta &= I_{\mathrm{vCLUB}}(\boldsymbol{X};\boldsymbol{L}) - I(\boldsymbol{X};\boldsymbol{L}) \\ &= \mathbb{E}_{P(\boldsymbol{X},\boldsymbol{L})}\left[\log Q_{\theta_X}(\boldsymbol{L}|\boldsymbol{X})\right] - \mathbb{E}_{P(\boldsymbol{X})}\mathbb{E}_{P(\boldsymbol{L})}\left[\log Q_{\theta_X}(\boldsymbol{L}|\boldsymbol{X})\right] \\ &\quad + \mathbb{E}_{P(\boldsymbol{X})}\mathbb{E}_{P(\boldsymbol{L})}\left[\log P(\boldsymbol{L})\right] - \mathbb{E}_{P(\boldsymbol{X},\boldsymbol{L})}\left[\log P(\boldsymbol{L}|\boldsymbol{X})\right] \\ &= \mathbb{E}_{P(\boldsymbol{X})}\mathbb{E}_{P(\boldsymbol{L})}\left[\log \frac{P(\boldsymbol{L})P(\boldsymbol{X})}{Q_{\theta_X}(\boldsymbol{L}|\boldsymbol{X})P(\boldsymbol{X})}\right] \\ &\quad - \mathbb{E}_{P(\boldsymbol{X},\boldsymbol{L})}\left[\log \frac{P(\boldsymbol{L}|\boldsymbol{X})P(\boldsymbol{X})}{Q_{\theta_X}(\boldsymbol{L}|\boldsymbol{X})P(\boldsymbol{X})}\right].\end{aligned} \tag{11}$$

Let $Q_{\theta_X}(\boldsymbol{L}|\boldsymbol{X})P(\boldsymbol{X}) = Q_{\theta_X}(\boldsymbol{X},\boldsymbol{L})$ be the induced variational joint distribution of $Q_{\theta_X}(\boldsymbol{L}|\boldsymbol{X})$. Then, the independence of $\boldsymbol{X}$ and $\boldsymbol{L}$ is a sufficient but not necessary condition for $I_{\mathrm{vCLUB}}(\boldsymbol{X};\boldsymbol{L}) = I(\boldsymbol{X};\boldsymbol{L})$. Thus

$$\begin{aligned}\Delta &= I_{\mathrm{vCLUB}}(\boldsymbol{X};\boldsymbol{L}) - I(\boldsymbol{X};\boldsymbol{L}) \\ &= D_{\mathrm{KL}}\left(P(\boldsymbol{X})P(\boldsymbol{L}) \,\|\, Q_{\theta_X}(\boldsymbol{L}|\boldsymbol{X})P(\boldsymbol{X})\right) \\ &\quad - D_{\mathrm{KL}}\left(P(\boldsymbol{X},\boldsymbol{L}) \,\|\, Q_{\theta_X}(\boldsymbol{L}|\boldsymbol{X})P(\boldsymbol{X})\right).\end{aligned} \tag{12}$$

Then, it follows that $I_{\mathrm{vCLUB}}(\boldsymbol{X};\boldsymbol{L})$ serves as an upper bound for $I(\boldsymbol{X};\boldsymbol{L})$ if and only if $D_{\mathrm{KL}}\big(P(\boldsymbol{X})P(\boldsymbol{L}) \,\|\, Q_{\theta_X}(\boldsymbol{L},\boldsymbol{X})\big) \geq D_{\mathrm{KL}}\big(P(\boldsymbol{X},\boldsymbol{L}) \,\|\, Q_{\theta_X}(\boldsymbol{L},\boldsymbol{X})\big)$. Furthermore, if $\boldsymbol{X}$ and $\boldsymbol{L}$ are independent, then $P(\boldsymbol{X})P(\boldsymbol{L}) = P(\boldsymbol{X},\boldsymbol{L})$.

Proof: The above expression indicates that $I_{\mathrm{vCLUB}}(\boldsymbol{X};\boldsymbol{L})$ serves as an upper bound for $I(\boldsymbol{X};\boldsymbol{L})$ if $P(\boldsymbol{X},\boldsymbol{L})$ is closer to the variational joint distribution $Q_{\theta_X}(\boldsymbol{L},\boldsymbol{X})$ than $P(\boldsymbol{X})P(\boldsymbol{L})$ is.

Therefore, it is necessary to minimize $D_{\mathrm{KL}}\big(P(\boldsymbol{X},\boldsymbol{L}) \,\|\, Q_{\theta_X}(\boldsymbol{L},\boldsymbol{X})\big)$

$$\begin{aligned}&\min_{\theta} D_{\mathrm{KL}}\left(P(\boldsymbol{X},\boldsymbol{L}) \,\|\, Q_{\theta_X}(\boldsymbol{L},\boldsymbol{X})\right) \\ &\quad = \min_{\theta}\left(\mathbb{E}_{P(\boldsymbol{X},\boldsymbol{L})}\left[\log P(\boldsymbol{L}|\boldsymbol{X})\right]\right. \\ &\qquad \left. - \mathbb{E}_{P(\boldsymbol{X},\boldsymbol{L})}\left[\log Q_{\theta_X}(\boldsymbol{L}|\boldsymbol{X})\right]\right).\end{aligned} \tag{13}$$

Since the first term contains no parameters θ, then $\min_{\theta}$ is equivalent to maximizing the log-likelihood of $Q_{\theta_X}(\boldsymbol{L}|\boldsymbol{X})$

$$\max_{\theta} \mathbb{E}_{P(\boldsymbol{X},\boldsymbol{L})}\left[\log Q_{\theta_X}(\boldsymbol{L}|\boldsymbol{X})\right]. \tag{14}$$

Thus, the log-likelihood function $\mathcal{L}(\theta) = 1/N \sum_{i=1}^{N} \log Q_{\theta_X}(\boldsymbol{L}_i|\boldsymbol{X}_i)$ can be maximized, and this constitutes an unbiased estimator of $\mathbb{E}_{P(\boldsymbol{X},\boldsymbol{L})}\big[\log Q_{\theta_X}(\boldsymbol{L}|\boldsymbol{X})\big]$.

Assuming the neural network Q_{θ_X} achieves an infinitesimally small approximation $D_{\mathrm{KL}}(P(\boldsymbol{L}|\boldsymbol{X})\|Q_{\theta_X}(\boldsymbol{L}|\boldsymbol{X})) \leq \varepsilon$ ($\varepsilon > 0$). Then as stated, $I_{\mathrm{vCLUB}}(\boldsymbol{X};\boldsymbol{L})$ is an upper bound for $I(\boldsymbol{X};\boldsymbol{L})$ when $D_{\mathrm{KL}}(P(\boldsymbol{X})P(\boldsymbol{L})\|Q_{\theta_X}(\boldsymbol{L},\boldsymbol{X})) \geq D_{\mathrm{KL}}(P(\boldsymbol{X},\boldsymbol{L})\|Q_{\theta_X}(\boldsymbol{L},\boldsymbol{X}))$.

Counter-Proof: If $D_{\mathrm{KL}}(P(\boldsymbol{L}|\boldsymbol{X})\|Q_{\theta_X}(\boldsymbol{L}|\boldsymbol{X})) \leq \varepsilon$, then

$$\begin{aligned}&D_{\mathrm{KL}}\left(P(\boldsymbol{X},\boldsymbol{L}) \,\|\, Q_{\theta_X}(\boldsymbol{X},\boldsymbol{L})\right) \\ &\quad = \mathbb{E}_{P(\boldsymbol{X},\boldsymbol{L})}\left[\log \frac{P(\boldsymbol{X},\boldsymbol{L})}{Q_{\theta_X}}, \boldsymbol{L}\right)\Bigg] \\ &\quad = \mathbb{E}_{P(\boldsymbol{X},\boldsymbol{L})}\left[\log \frac{P(\boldsymbol{L}|\boldsymbol{X})}{Q_{\theta_X}(\boldsymbol{L}|\boldsymbol{X})}\right] \\ &\quad = D_{\mathrm{KL}}\left(P(\boldsymbol{L}|\boldsymbol{X}) \,\|\, Q_{\theta_X}(\boldsymbol{L}|\boldsymbol{X})\right) \leq \varepsilon.\end{aligned} \tag{15}$$

If $D_{\mathrm{KL}}\big(P(\boldsymbol{X})P(\boldsymbol{L}) \,\|\, Q_{\theta_X}(\boldsymbol{L},\boldsymbol{X})\big) < D_{\mathrm{KL}}\big(P(\boldsymbol{X},\boldsymbol{L}) \,\|\, Q_{\theta_X}(\boldsymbol{L},\boldsymbol{X})\big)$, then $0 < D_{\mathrm{KL}}\big(P(\boldsymbol{X})P(\boldsymbol{L}) \,\|\, Q_{\theta_X}(\boldsymbol{L},\boldsymbol{X})\big) < \varepsilon$. At this point

$$\begin{aligned}&|I_{\mathrm{vCLUB}}(\boldsymbol{X};\boldsymbol{L}) - I(\boldsymbol{X};\boldsymbol{L})| \\ &\quad = |D_{\mathrm{KL}}\left(P(\boldsymbol{X})P(\boldsymbol{L}) \,\|\, Q_{\theta_X}(\boldsymbol{L},\boldsymbol{X})\right) \\ &\qquad - D_{\mathrm{KL}}\left(P(\boldsymbol{X},\boldsymbol{L}) \,\|\, Q_{\theta_X}(\boldsymbol{L},\boldsymbol{X})\right)| \\ &\quad < \max\left\{D_{\mathrm{KL}}\left(P(\boldsymbol{X})P(\boldsymbol{L}) \,\|\, Q_{\theta_X}(\boldsymbol{L},\boldsymbol{X})\right),\right. \\ &\qquad \left. D_{\mathrm{KL}}\left(P(\boldsymbol{X},\boldsymbol{L}) \,\|\, Q_{\theta_X}(\boldsymbol{L},\boldsymbol{X})\right)\right\} \\ &\quad \leq \varepsilon.\end{aligned} \tag{16}$$

In summary, I_{vCLUB} is either an upper bound for mutual information or an estimator of mutual information with absolute error bounded by $D_{\mathrm{KL}}(P(\boldsymbol{L}|\boldsymbol{X})\|Q_{\theta_X}(\boldsymbol{L}|\boldsymbol{X}))$ infinitesimally small. Then, $I_{\mathrm{vCLUB}}(\boldsymbol{X};\boldsymbol{L})$ can be approximated as an upper bound for $I(\boldsymbol{X};\boldsymbol{L})$.

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3) Decoupling Term: For the disentanglement term: When directly optimizing $I(\boldsymbol{Y};\boldsymbol{H})$, the corresponding variational approximation $Q_\theta(\boldsymbol{H}|\boldsymbol{Y})$ cannot yield a directly computable upper bound. This is because minimizing this term may lead to erasure of label-relevant information in $\boldsymbol{H}$, only the data distribution $P(\boldsymbol{Y})$ is learned without encoding label-independent information into the sample representation $\boldsymbol{H}$, thereby failing to achieve decoupling between variables $\boldsymbol{H}$ and $\boldsymbol{L}$.

a) Proof that $P(\boldsymbol{Y})$ is a trivial solution of $Q_\theta(\boldsymbol{H}|\boldsymbol{Y})$: Let $Q_\theta(\boldsymbol{H}|\boldsymbol{Y}) = P(\boldsymbol{Y})$. Since $P(\boldsymbol{Y})$ is the label distribution of the dataset, sampling via the Dirac delta function yields $P(\boldsymbol{Y} = y) = \sum_{i=1}^{N} \delta(x - x_i)p(y|x_i)$, where $\delta(\cdot)$ denotes the dirac delta function.

For the original optimization problem, the computable upper bound of $I(\boldsymbol{Y};\boldsymbol{H})$ is

$$\begin{aligned} &I_{\text{vCLUB}}(\boldsymbol{Y};\boldsymbol{H}) \\ &\quad = \mathbb{E}_{P(\boldsymbol{Y},\boldsymbol{H})}\left[\log P(\boldsymbol{Y})\right] - \mathbb{E}_{P(\boldsymbol{Y})}\mathbb{E}_{P(\boldsymbol{H})}\left[\log P(\boldsymbol{Y})\right] \\ &\quad = 0. \end{aligned} \tag{17}$$

At this point, $I(\boldsymbol{Y};\boldsymbol{H})$ has been minimized, but $\boldsymbol{H}$ remains unoptimized, constituting a trivial solution.

b) Problem transformation: Conditional independence assumption: $P(\boldsymbol{X},\boldsymbol{Y},\boldsymbol{H}) = P(\boldsymbol{X})P(\boldsymbol{Y}|\boldsymbol{X})P(\boldsymbol{H}|\boldsymbol{X})$, assuming $\boldsymbol{Y}$ and $\boldsymbol{H}$ are independent given $\boldsymbol{X}$. This implies: $P(\boldsymbol{Y},\boldsymbol{H}|\boldsymbol{X}) = P(\boldsymbol{Y}|\boldsymbol{X})P(\boldsymbol{H}|\boldsymbol{X})$ while $P(\boldsymbol{Y},\boldsymbol{H}) \neq P(\boldsymbol{Y})P(\boldsymbol{H})$. For the mutual information $I(\boldsymbol{X};\boldsymbol{Y},\boldsymbol{H})$

$$\begin{aligned} I(\boldsymbol{X};\boldsymbol{Y},\boldsymbol{H}) &= \mathbb{E}_{P(\boldsymbol{X},\boldsymbol{Y},\boldsymbol{H})}\left[\log \frac{P(\boldsymbol{X},\boldsymbol{Y},\boldsymbol{H})}{P(\boldsymbol{X})P(\boldsymbol{Y},\boldsymbol{H})}\right] \\ &= \mathbb{E}_{P(\boldsymbol{X},\boldsymbol{Y},\boldsymbol{H})}\left[\log \frac{P(\boldsymbol{Y}|\boldsymbol{X})}{P(\boldsymbol{Y})} + \log \frac{P(\boldsymbol{H}|\boldsymbol{X})}{P(\boldsymbol{H})} \right. \\ &\qquad \left. + \log \frac{P(\boldsymbol{Y})P(\boldsymbol{H})}{P(\boldsymbol{Y},\boldsymbol{H})}\right] \\ &= I(\boldsymbol{X};\boldsymbol{Y}) + I(\boldsymbol{X};\boldsymbol{H}) - I(\boldsymbol{Y};\boldsymbol{H}). \end{aligned} \tag{18}$$

We obtain: $I(\boldsymbol{Y};\boldsymbol{H}) = I(\boldsymbol{X};\boldsymbol{Y}) + I(\boldsymbol{X};\boldsymbol{H}) - I(\boldsymbol{X};\boldsymbol{Y},\boldsymbol{H})$ where $I(\boldsymbol{Y};\boldsymbol{H}) = I(\boldsymbol{X};\boldsymbol{Y}) + I(\boldsymbol{X};\boldsymbol{H}) - I(\boldsymbol{X};\boldsymbol{Y},\boldsymbol{H})$ contains no optimization terms.

For $I(\boldsymbol{X};\boldsymbol{H})$, an upper bound can be constructed using the variational distribution $Q_{\theta_{\boldsymbol{H}}}(\boldsymbol{H}|\boldsymbol{X})$

$$\begin{aligned} I(\boldsymbol{X};\boldsymbol{H}) &\leq I_{\text{vCLUB}}(\boldsymbol{X};\boldsymbol{H}) \\ &= \mathbb{E}_{P(\boldsymbol{X},\boldsymbol{H})}\left[\log Q_{\theta_{\boldsymbol{H}}}(\boldsymbol{H}|\boldsymbol{X})\right] \\ &\quad - \mathbb{E}_{P(\boldsymbol{X})}\mathbb{E}_{P(\boldsymbol{H})}\left[\log Q_{\theta_{\boldsymbol{H}}}(\boldsymbol{H}|\boldsymbol{X})\right]. \end{aligned} \tag{19}$$

For $I(\boldsymbol{X};\boldsymbol{Y},\boldsymbol{H})$, a lower bound can be constructed utilizing the variational distribution $Q_{\phi_{\boldsymbol{H}}}(\boldsymbol{X}|\boldsymbol{Y},\boldsymbol{H})$

$$\begin{aligned} &I(\boldsymbol{X};\boldsymbol{Y},\boldsymbol{H}) \\ &\quad \geq \mathbb{E}_{P(\boldsymbol{X},\boldsymbol{Y},\boldsymbol{H})}\left[\log Q_{\phi_{\boldsymbol{H}}}(\boldsymbol{X}|\boldsymbol{Y},\boldsymbol{H})\right] + H(\boldsymbol{X}). \end{aligned} \tag{20}$$

D. Patch-Mix-Based Virtual Domain Generation

To generalize knowledge decoupling capabilities from the source domain to unseen target domains for classification tasks, two core objectives must be addressed.

1) *Domain-Invariant Feature Learning:* Extract universal features correlated solely with land-cover classes, mitigating distribution gaps between source and target domains.
2) *Cross-Domain Robustness Optimization:* Capture source-specific features by expanding decision boundaries to enhance robustness against DS. These interdependent objectives risk optimization conflicts: Overemphasis on source features may cause overfitting, while excessive boundary expansion degrades class separability—both leading to suboptimal performance in target domains.

Therefore, this work constructs a discriminator $D(\cdot)$ to capture domain-invariant feature representations and a generator $G(\cdot)$ for virtual domain generation. Through adversarial optimization between $D(\cdot)$ and $G(\cdot)$, these components jointly achieve both optimization objectives.

1) Spectral–Spatial Joint Encoder: The spectral–spatial joint encoder employs multiscale convolution and attention mechanisms to model spectral–spatial interactions for the spectral–spatial invariant feature $\boldsymbol{L}$ extracted via knowledge disentanglement, generating discriminative spectral–spatial joint features $\boldsymbol{Z}_{\text{spe-spa}}$. These features provide cross-domain invariant discriminative information for virtual domain generation, ensuring the separability of virtual domains.

2) Patch-Mix-Based Feature Perturbation: Domain-sensitive features $\boldsymbol{H}$ carry heterogeneous perturbations uncorrelated with land-cover classes. These features exhibit significant distribution shifts across domains, causing pronounced DS that hinder effective cross-scene HSI classification. Applying controlled perturbations to such features expands decision boundaries, thereby enhancing robustness in cross-scene classification performance.

Current feature perturbation methods like adaptive instance normalization (AdaIN) typically perturb entire feature maps per channel. This global perspective overlooks inherent spatial heterogeneity and spectral continuity of HSI, limiting localized feature exploration. Therefore, we propose the Patch-Mix-based perturbation method that divides feature maps into patches and applies perturbations at the patch level. This enhances variability in feature statistics (mean/variance), significantly improving statistical diversity.

First, domain-sensitive features $\boldsymbol{H}$ are spatially partitioned into nonoverlapping patches $\boldsymbol{H}^P$, where P denotes the patch count, and H_p, W_p represent height and width of each patch. The mean μ_b^p and standard deviation σ_b^p statistics for the pth patch at bth spectral band are computed as

$$\begin{cases} \mu_b^p = \dfrac{1}{H_p W_p}\displaystyle\sum_{h=1}^{H_p}\sum_{w=1}^{W_p}\boldsymbol{H}_b^p \\ \sigma_b^p = \sqrt{\dfrac{1}{H_p W_p}\displaystyle\sum\nolimits_{h=1}^{H_p}\sum\nolimits_{w=1}^{W_p}\left(\boldsymbol{H}_b^p - \mu_b^p\right)^2}. \end{cases} \tag{21}$$

Then, the bth patch statistics are computed as defined above

$$\begin{cases} \mu_b = \left(\mu_b^1, \mu_b^2, \mu_b^3, \ldots, \mu_b^p\right) \\ \sigma_b = \left(\sigma_b^1, \sigma_b^2, \sigma_b^3, \ldots, \sigma_b^p\right). \end{cases} \tag{22}$$

For the statistics of bth spectral band, randomly reshuffle the sequences of μ^p and σ^p to obtain perturbed statistics $\tilde{\mu}_b$ and $\tilde{\sigma}_b$

$$\begin{cases} \tilde{\mu}_b = \left(\tilde{\mu}_b^1, \tilde{\mu}_b^2, \tilde{\mu}_b^3, \ldots, \tilde{\mu}_b^p\right) \\ \tilde{\sigma}_b = \left(\tilde{\sigma}_b^1, \tilde{\sigma}_b^2, \tilde{\sigma}_b^3, \ldots, \tilde{\sigma}_b^p\right). \end{cases} \tag{23}$$

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TABLE I
NUMBER OF CLASSES AND SAMPLES OF PAVIA DATASET

Class			Number of Samples	
No.	Color	Name	PU	PC
P1		Tree	3064	7598
P2		Asphalt	6631	9248
P3		Brick	3682	2685
P4		Bitumen	1330	7287
P5		Shadow	947	2863
P6		Meadow	18649	3090
P7		Bare soil	5029	6584
Total			39332	39355

Next, the variance of these perturbed statistics for patch features H^p is expressed as

$$\begin{cases} \Sigma^2_{\tilde{\mu}}\left(\boldsymbol{H}^p\right) = \dfrac{1}{P}\displaystyle\sum_{p=1}^{P}\left(\tilde{\mu}\left(\boldsymbol{H}^p\right) - E_p\left[\tilde{\mu}\left(\boldsymbol{H}^p\right)\right]\right)^2 \\ \Sigma^2_{\tilde{\sigma}}\left(\boldsymbol{H}^p\right) = \dfrac{1}{P}\displaystyle\sum_{p=1}^{P}\left(\tilde{\sigma}\left(\boldsymbol{H}^p\right) - E_p\left[\tilde{\sigma}\left(\boldsymbol{H}^p\right)\right]\right)^2 \end{cases} \tag{24}$$

where $\Sigma^2_{\tilde{\mu}} \in \mathbb{R}^C$ and $\Sigma^2_{\tilde{\sigma}} \in \mathbb{R}^C$ represent uncertainty estimates for the reshuffled statistics $\tilde{\mu}(\boldsymbol{H}^p)$ and $\tilde{\sigma}(\boldsymbol{H}^p)$ of patch features $\boldsymbol{H}^p$, respectively.

After obtaining the variance of $\boldsymbol{H}^p$, diverse new feature statistics are generated through Gaussian perturbation

$$\begin{cases} \hat{\mu}\left(\boldsymbol{H}^p\right) = \tilde{\mu}\left(\boldsymbol{H}^p\right) + \Sigma_{\tilde{\mu}}\left(\boldsymbol{H}^p\right)\cdot\varepsilon_{\tilde{\mu}}, \quad \varepsilon_{\tilde{\mu}} \sim \mathcal{N}(0,1) \\ \hat{\sigma}\left(\boldsymbol{H}^p\right) = \tilde{\sigma}\left(\boldsymbol{H}^p\right) + \Sigma_{\tilde{\sigma}}\left(\boldsymbol{H}^p\right)\cdot\varepsilon_{\tilde{\sigma}}, \quad \varepsilon_{\tilde{\sigma}} \sim \mathcal{N}(0,1). \end{cases} \tag{25}$$

Finally, the generated mixed features $\boldsymbol{Z}_{\text{Mix}}$ are obtained by replacing original patch feature statistics with perturbed values using the AdaIN method

$$\boldsymbol{Z}_{\text{Mix}} = \hat{\sigma}\left(\boldsymbol{H}^p\right)\cdot\frac{\boldsymbol{H}^p - \mu\left(\boldsymbol{H}^p\right)}{\sigma\left(\boldsymbol{H}^p\right)} + \hat{\mu}\left(\boldsymbol{H}^p\right). \tag{26}$$

3) Virtual Domain Generation: Using the spectral–spatial joint features $\boldsymbol{Z}_{\text{spe-spa}}$ and mixed features $\boldsymbol{Z}_{\text{Mix}}$, virtual domains $\tilde{\boldsymbol{X}}$ are generated as follows:

$$\tilde{\boldsymbol{X}} = \Gamma\left(\text{CONCAT}\left(\boldsymbol{Z}_{\text{spe-spa}}, \boldsymbol{Z}_{\text{Mix}}\right)\right) \tag{27}$$

where $\Gamma(\cdot)$ incorporates convolutional and activation operations to restore dimensions consistent with $\boldsymbol{X}$.

E. Adversarial Optimization Learning

To mitigate model optimization difficulty, the intermediate domain $\hat{\boldsymbol{X}}$ is generated by random linear combination of source domain data $\boldsymbol{X}$ and virtual domain $\tilde{\boldsymbol{X}}$

$$\hat{\boldsymbol{X}} = \gamma\tilde{\boldsymbol{X}} + (1-\gamma)\boldsymbol{X} \tag{28}$$

where γ is a random coefficient sampled from a continuous uniform distribution over $[0, 1]$.

1) Discriminator Loss: The discriminator comprises a feature extractor $f(\cdot)$, a classification head $C(\cdot)$, and a projection head $P(\cdot)$: $f(\cdot)$ learns cross-domain invariant features, $C(\cdot)$ produces classified outputs, and $P(\cdot)$ outputs projected features for supervised contrastive and adversarial learning.

The classification task employs cross-entropy loss, formally expressed as

$$\mathcal{L}_{\text{CE}}\left(p_i, y_i\right) = -\sum_c y_i^c \log p_i^c \tag{29}$$

where p_i denotes the predicted probability vector from classifier $C(\cdot)$, y_i is the one-hot encoded ground truth label, and c indexes the class. The classification loss on $\boldsymbol{X}$, $\tilde{\boldsymbol{X}}$, and $\hat{\boldsymbol{X}}$ is

$$\begin{cases} \mathcal{L}_{\boldsymbol{X}} = \dfrac{1}{N}\displaystyle\sum_{i=1}^{N}\mathcal{L}_{\text{CE}}\left(C\left(\boldsymbol{X}_i\right), y_i\right) \\ \mathcal{L}_{\tilde{\boldsymbol{X}}} = \dfrac{1}{N}\displaystyle\sum_{i=1}^{N}\mathcal{L}_{\text{CE}}\left(C\left(\tilde{\boldsymbol{X}}_i\right), y_i\right) \\ \mathcal{L}_{\hat{\boldsymbol{X}}} = \dfrac{1}{N}\displaystyle\sum_{i=1}^{N}\mathcal{L}_{\text{CE}}\left(C\left(\hat{\boldsymbol{X}}_i\right), y_i\right) \end{cases} \tag{30}$$

where N denotes the number of samples.

These classification losses reflect the performance of the classifier at various stages of cross-scene learning. To enhance learning of class-shared features from same-class samples across different domains, an intraclass supervised contrastive loss is introduced

$$\mathcal{L}_{\text{con}} = \sum_i^N -\log\left(\frac{\exp\left(z_i\cdot z_i^+\right)}{\sum_{j=1,j\neq i}^N \exp\left(z_i\cdot z_j\right)}\right) \tag{31}$$

where z represents the projected features, and z^+ denotes positive sample features (i.e., features from the same-class samples). Through optimization of $\mathcal{L}_{\text{con}}$, features of the same-class samples are pulled closer while different-class features are pushed apart.

The final optimization objective of discriminator jointly optimizes the feature extractor $(\cdot;\boldsymbol{\theta}_f)$, classification head $(\cdot;\boldsymbol{\theta}_C)$, and projection head $(\cdot;\boldsymbol{\theta}_P)$ through a combination of classification loss and intraclass supervised contrastive loss

$$\min_{\theta_f,\theta_C,\theta_P}\mathcal{L} = \mathcal{L}_{\boldsymbol{X}} + \mathcal{L}_{\tilde{\boldsymbol{X}}} + \mathcal{L}_{\hat{\boldsymbol{X}}} + \lambda_1\mathcal{L}_{\text{con}} \tag{32}$$

where λ_1 is a hyperparameter employed to govern the tradeoff of the supervised intraclass contrastive learning loss.

2) Generator Loss: The generator $G(\cdot)$ aims to generate samples with maximized divergence, inducing distinct feature distributions between the generated virtual domain and source domain. This strategy expands decision boundaries, which motivates the introduction of the interclass adversarial loss

$$\mathcal{L}_{\text{adv}} = \sum_i^N \log\left(1 - \frac{\exp\left(z_i\cdot z_i^+\right)}{\sum_{j=1,j\neq i}^N \exp\left(z_i\cdot z_j\right)}\right). \tag{33}$$

Furthermore, inspired by [40], this work constructs a generative similarity loss that models fine-grained discrepancies between source and virtual domains from dual perspectives of structural and textural similarity.

The structural similarity loss can be expressed as

$$\mathcal{L}_{\text{sim}} = \sum_{w=0}^{W} r_w\cdot\left(1 - \text{sim}\left(\boldsymbol{X}^w, \tilde{\boldsymbol{X}}^w\right)\right) \tag{34}$$

where $\text{sim}(\cdot)$ denotes the structural similarity operator, $\boldsymbol{X}^w$ and $\tilde{\boldsymbol{X}}^w$ represent the corresponding frequency components,

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$w = \{1, 2, 3, 4\}$ is the index of the respective frequency component $\{\mathrm{LL}, \mathrm{LH}, \mathrm{HL}, \mathrm{LL}\}$, and r_w denotes the weight assigned to the corresponding component

$$\mathrm{LL} : \mathrm{LH} : \mathrm{HL} : \mathrm{HH} = r^2 : r(1-r) : r(1-r) : (1-r)^2. \quad (35)$$

The textural similarity loss can be expressed as

$$\mathcal{L}_{\mathrm{grad}} = \sum_{w=0}^{W} r_w \left\| |\nabla \boldsymbol{X}^w| - \left|\nabla \tilde{\boldsymbol{X}}^w\right| \right\|_1 \quad (36)$$

where $\|\cdot\|_1$ denotes the $L1$-norm operator, and ∇ corresponds to the sobel gradient operator.

The ultimate optimization objective of $G(\cdot)$ is achieved by jointly optimizing $(\cdot; \boldsymbol{\theta}_G)$ through both the interclass adversarial loss and the generative similarity loss, with the additional application of classification loss of $\tilde{\boldsymbol{X}}$ to ensure $G(\cdot)$ operates within the correct label space

$$\min_{\theta_G} \mathcal{L} = \mathcal{L}_{\mathrm{ED}} + \mathcal{L}_{\mathrm{sim}} + \mathcal{L}_{\mathrm{grad}} + \lambda_2 \mathcal{L}_{\mathrm{adv}}. \quad (37)$$

IV. Experiment and Analysis

A. Data Description

To validate the efficacy of SIBDNet for cross-scene HSI classification under single-source DG, experiments were conducted on three cross-scene HSI datasets: the Pavia dataset, Houston dataset, and HyRank dataset. Section IV-A1–IV-A3 provides detailed descriptions of these datasets.

1) Pavia Dataset: This dataset comprises two subsets: the Pavia University (PU) and Pavia Center (PC) datasets. The PU dataset has dimensions of 610 × 340 pixels with 103 spectral bands, while the PC dataset measures 1096 × 715 pixels and contains 102 spectral bands. To ensure spectral band consistency between the subsets, we followed the experimental protocol in [13] by removing the terminal spectral band from the PU dataset. The seven shared land-cover classes are cataloged in Table I. PU is the source domain, and PC is the target domain.

2) Houston Dataset: This study utilizes HSI datasets acquired over the University of Houston campus and surrounding areas of Houston13 and Houston18. The Houston13 dataset comprises 144 spectral bands with a spatial resolution of 2.5 m, while Houston18 contains 48 spectral bands at 1-m spatial resolution. Following the experimental protocol in [13], we selected a 210×954 pixel region from corresponding scenes in both datasets. To ensure spectral band consistency between source and target domains, only the 48 bands of Houston13 were employed during training. The seven shared land-cover classes are documented in Table II. Houston13 is the source domain, and Houston18 is the target domain.

3) HyRank Dataset: This dataset comprises two subsets: the Dioni and Loukia datasets, both acquired by the Earth Observing-1 satellite sensor operated by the United States Geological Survey. Each scene features a 30-m spatial resolution with 176 spectral channels. The Dioni subset has dimensions of 250 × 1376 pixels, while Loukia measures 249 × 945 pixels. The twelve shared land-cover classes are cataloged in Table III. Dioni is the source domain, and Loukia is the target domain.

TABLE II
Number of Classes and Samples of Houston Dataset

Class			Number of Samples	
No.	Color	Name	Houston13	Houston18
H1		Grass Healthy	345	1353
H2		Grass Stressed	365	4888
H3		Trees	365	2766
H4		Water	285	22
H5		Residential Buildings	319	5347
H6		Non-residential Buildings	480	6365
H7		Road	443	6365
Total			2530	53200

TABLE III
Number of Classes and Samples of HyRank Dataset

Class			Number of Samples	
No.	Color	Name	Dioni	Loukia
H1		Dense Urban Fabric	1262	206
H2		Mineral Extraction Sites	204	54
H3		Non lrrigated Arable Land	614	426
H4		Fruit Trees	150	79
H5		Olive Groves	1768	1107
H6		Coniferous Forest	361	422
H7		Dense Sderophyllous Vegetation	5035	2996
H8		Sparce Sderophyllous Vegetation	6374	2361
H9		Sparcely Vegetated Areas	1754	399
H10		Rocks and Sand	492	453
H11		Water	1612	1393
H12		Coastal Water	398	421
Total			20024	10317

B. Experimental Setting

All experiments were conducted on a workstation equipped with an AMD Ryzen 9 9950X CPU (5.70 GHz), 64-GB RAM, and an NVIDIA GeForce RTX 4080 SUPER GPU, utilizing the PyTorch framework. The comparative methods in this study include noncross-domain method: dual-branch subpixel-guided network (DSNet) [8]; and DG methods: SDEnet [13], LDGnet [15], LLURnet [29], FDGNet [14], invariant semantic DG shuffle network (ISDGS) [28], and adversarial decoupling domain generalization network (ADNet) [41]. To ensure fair comparison, all experiments adhered to the protocol where the target domain remains unseen during training, models were exclusively trained on source domain data and evaluated on target domain data. For SIBDNet training, we employed the Adam optimizer with 200 epochs, a learning rate of 0.001, and a patch size of 13 × 13 pixels.

Three evaluation metrics were selected, comprising overall accuracy (OA), average accuracy (AA) per class, and the Kappa coefficient.

C. Comparison Results Analysis

The experimental results of SIBDNet and comparative methods across the three datasets are presented in Tables IV–VI, from which the following observations can be drawn.

1) Compared to the noncross-domain method DSNet, SIBDNet achieves superior classification accuracy. This stems from the following: noncross-domain methods adopt an empirical risk minimization learning paradigm predicated on the assumption that training and test

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TABLE IV

CLASSIFICATION PERFORMANCE OF DIFFERENT METHODS (PAVIA DATASET)

Class	DSNet	SDEnet	LDGnet	LLURnet
P1	92.66±1.99	84.71±4.61	**93.41±2.76**	86.25±5.20
P2	80.63±8.80	80.45±2.59	83.75±2.54	84.41±1.22
P3	35.20±9.08	68.76±8.25	64.21±9.55	68.26±5.73
P4	37.68±6.92	80.87±5.58	79.13±3.39	79.68±2.42
P5	**98.37±0.55**	87.97±4.81	85.12±4.32	85.10±3.99
P6	**85.70±2.37**	80.41±4.57	69.95±4.72	73.17±5.18
P7	51.88±7.94	76.12±4.22	79.03±7.35	79.02±2.94
OA(%)	68.78±2.05	80.37±0.84	81.65±0.89	81.05±0.65
AA(%)	68.85±1.69	79.90±1.46	79.23±1.30	79.41±1.06
Kappa(%)	62.50±2.46	76.49±1.03	77.87±1.04	77.22±0.78
Class	FDGNet	ISDGS	ADNet	SIBDNet
P1	87.58±3.10	86.57±5.29	87.27±3.50	90.18±5.62
P2	81.55±2.63	87.24±2.91	87.19±4.63	**87.56±2.87**
P3	75.94±4.98	63.81±5.99	65.0±12.03	**79.69±9.70**
P4	**84.12±2.73**	78.42±7.29	83.15±5.46	81.60±2.66
P5	87.62±3.30	89.73±6.02	85.75±4.32	92.38±2.46
P6	74.53±2.78	77.14±4.21	75.62±2.61	71.95±5.38
P7	80.84±2.75	78.16±6.16	78.75±6.95	**85.38±6.10**
OA(%)	82.58±1.10	81.74±1.36	82.52±0.58	**85.18±0.76**
AA(%)	81.74±1.30	80.15±1.54	80.40±1.35	**84.11±1.16**
Kappa(%)	79.10±1.32	78.0±1.64	78.90±0.69	**82.20±0.90**

TABLE V

CLASSIFICATION PERFORMANCE OF DIFFERENT METHODS (HOUSTON DATASET)

Class	DSNet	SDEnet	LDGnet	LLURnet
H1	**97.36±1.43**	43.07±8.44	67.82±8.68	54.80±8.89
H2	5.79±5.69	75.10±5.92	77.32±4.36	80.68±8.00
H3	7.72±4.60	59.45±4.38	62.25±5.05	62.65±4.27
H4	5.20±1.59	**100±0**	96.59±6.72	**100±0**
H5	50.25±6.79	66.42±3.15	72.74±7.81	70.37±4.14
H6	74.50±8.90	88.45±2.21	**92.53±1.83**	86.28±1.79
H7	51.29±15.38	52.09±7.28	38.24±3.57	59.09±2.88
OA(%)	60.05±2.98	78.00±0.52	80.44±0.69	78.89±0.39
AA(%)	41.73±2.69	69.22±2.37	72.48±1.60	73.41±0.72
Kappa(%)	34.53±2.49	62.28±1.32	65.80±1.20	64.71±0.62
Class	FDGNet	ISDGS	ADNet	SIBDNet
H1	38.35±7.16	48.16±9.27	40.06±11.29	61.48±7.84
H2	81.87±3.21	76.69±5.27	79.00±6.88	**85.12±3.23**
H3	62.27±3.18	65.26±3.15	60.59±5.52	**68.18±5.11**
H4	**100±0**	**100±0**	**100±0**	**100±0**
H5	72.61±7.87	65.42±10.46	68.10±7.62	66.97±5.77
H6	88.76±1.92	90.89±2.48	90.94±1.42	91.53±1.50
H7	56.64±4.97	50.40±2.86	53.63±6.05	**61.64±4.53**
OA(%)	80.00±0.42	79.76±0.84	80.22±0.45	**82.93±0.57**
AA(%)	71.50±1.49	70.98±0.71	70.33±1.84	**76.46±1.13**
Kappa(%)	66.00±0.77	64.53±0.97	65.74±1.09	**71.02±1.77**

TABLE VI

CLASSIFICATION PERFORMANCE OF DIFFERENT METHODS (HYRANK DATASET)

Class	DSNet	SDEnet	LDGnet	LLURnet
D1	0.73±0.61	20.71±1.41	43.98±6.27	37.58±5.48
D2	28.40±12.50	43.21±7.46	2.59±2.77	0.46±1.10
D3	2.62±3.82	37.53±7.55	**56.81±7.19**	46.77±5.43
D4	0.85±0.94	6.47±4.07	20.76±8.87	0.95±1.64
D5	0±0	1.91±2.64	0.05±0.11	0.01±0.03
D6	18.84±5.36	**46.13±4.05**	38.72±1.84	43.40±3.40
D7	**93.04±3.02**	76.97±2.07	80.60±3.54	74.29±3.30
D8	1.79±1.23	57.19±4.67	48.23±4.34	70.52±3.54
D9	8.69±10.55	**75.19±2.56**	74.49±6.00	33.87±5.49
D10	**73.36±4.10**	0.37±1.04	0.31±0.51	0.02±0.06
D11	**100±0**	**100±0**	**100±0**	**100±0**
D12	94.62±5.02	**100±0**	**100±0**	**100±0**
OA(%)	49.40±0.75	60.28±1.07	59.97±0.38	61.07±0.52
AA(%)	35.24±1.96	47.14±1.18	46.67±1.21	42.31±0.86
Kappa(%)	38.21±0.71	51.04±1.18	50.80±0.34	51.62±0.61
Class	FDGNet	ISDGS	ADNet	SIBDNet
D1	23.98±7.59	**47.67±7.39**	30.63±13.38	24.15±6.31
D2	**72.22±9.66**	21.48±12.92	16.67±19.39	57.87±12.99
D3	49.86±4.16	55.70±10.98	54.55±8.92	53.40±9.96
D4	12.91±6.57	**28.74±14.50**	16.71±20.84	10.13±2.68
D5	1.45±1.42	3.69±5.32	1.07±0.81	**17.39±3.38**
D6	34.74±1.90	24.48±12.73	32.47±10.73	19.91±8.94
D7	76.13±1.79	75.28±2.80	76.66±3.87	80.69±2.97
D8	**73.55±3.70**	63.82±6.14	72.42±4.65	68.70±2.09
D9	47.67±7.44	64.86±17.84	40.98±10.46	63.54±2.74
D10	0±0	8.68±17.68	1.04±3.11	0.08±0.14
D11	**100±0**	**100±0**	**100±0**	**100±0**
D12	**100±0**	**100±0**	**100±0**	**100±0**
OA(%)	62.92±0.27	61.92±0.72	62.57±0.48	**64.93±0.41**
AA(%)	49.38±0.87	49.53±1.72	45.27±3.05	**49.63±1.26**
Kappa(%)	53.91±0.35	53.29±0.83	53.43±0.60	**56.44±0.45**

sets are independently and identically distributed. However, this framework faces DS challenges in cross-scene scenes, resulting in models that perform well on the source domain but exhibit limited generalization to the target domain. In contrast, SIBDNet overcomes DS challenges, thereby enabling generalization to unseen target domains.

2) Compared to the DG method SDEnet, SIBDNet achieves OA improvements of 3.00%, 4.09%, and 3.01% across the three datasets, respectively. This performance advantage originates from the following distinction. SDEnet treats morphological structure information in datasets as a physically driven prior constraint. Different HSIs often exhibit significant variations in spatial organization and morphological features of corresponding land-cover classes due to divergent acquisition regions. In contrast, SIBDNet does not presume dependence on morphological priors. Instead, it employs spectrum decomposition and knowledge disentanglement to capture transferable invariant spectral–spatial features from the source domain, thereby learning robust DS mitigation strategies.
3) Compared to the DG method LDGnet, SIBDNet achieves OA improvements of 2.62%, 1.74%, and 4.16% across the three datasets, respectively. This performance differential arises from fundamental methodological distinctions. While LDGnet constructs DS-robust feature spaces through text–visual modality alignment, the effectiveness of such modality-aligned features critically depends on meticulously engineered textual descriptors. In contrast, SIBDNet employs information-bottleneck knowledge disentanglement to extract class-discriminative spectral–spatial invariant features. This approach neither relies on external auxiliary information nor requires handcrafted annotations, deriving representations solely from inherent spectral and structural features within HSI data itself. Furthermore, it strategically perturbs domain-sensitive features to mitigate DS challenges.
4) Compared to the DG method LLURnet, SIBDNet achieves OA improvements of 2.06%, 3.11%, and 3.36% across the three datasets, respectively. This performance differential stems from core architectural distinctions. While LLURnet eliminates style bias by generating

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unbiased style cube collections, this global random augmentation method lacks explicit disentanglement mechanisms for domain-invariant and domain-sensitive features. Consequently, it inadvertently corrupts domain-invariant representations, resulting in non-negligible performance degradation. In contrast, SIBDNet employs spectrum decomposition and information bottleneck disentanglement to separate spectral–spatial invariant features from domain-sensitive features. Furthermore, it deliberately designs a Patch-Mix feature perturbation strategy for domain-sensitive features, thereby mitigating DS while preserving the integrity of invariant representations.

5) Compared to the DG method, FDGNet, and SIBDNet achieve OA improvements of 1.56%, 2.33%, and 1.86% across the three datasets, respectively. This performance advantage arises from fundamental representational distinctions. FDGNet leverages data geometric manifolds as reliability constraints and employs a class-wise aware manifold sampling strategy to generate intermediate subdomains. However, in complex scenes, such data manifolds fail to accurately characterize feature distributions, consequently impairing the model's ability to fully capture class-specific spectral–spatial correspondences and resulting in suboptimal performance. In contrast, SIBDNet utilizes structurally simplified filters conforming to spectral physical properties to decouple high-frequency and low-frequency features in HSI data. These features are subsequently refined through information-bottleneck knowledge disentanglement, enabling comprehensive learning of the inherent mappings between inherently coupled spectral–spatial features and land-cover classes, thereby enhancing model robustness in complex scenes.
6) Compared to the DG method ISDGS, SIBDNet achieves OA improvements of 3.42%, 3.17%, and 3.01% across the three datasets, respectively. The reason is that while ISDGS introduces innovations in feature style control and spatial structure perturbation, its generation process lacks guidance mechanisms, limiting its effectiveness under complex DSs. In contrast, SIBDNet, through 3-D DWT decomposition and the information bottleneck mechanism, achieves precise separation of spectral–spatial invariant features from domain-sensitive features. Building upon this disentangled structure, our Patch-Mix method enables controlled perturbation of domain-sensitive features, effectively expanding the decision boundary while preserving the integrity of classification features. This results in stronger generalization capability under complex DSs.
7) Compared to the DG method, ADNet, and SIBDNet achieve OA improvements of 2.66%, 2.71%, and 2.36% across the three datasets, respectively. The reason is: ADNet's adversarial disentanglement of HSI into causal and noncausal components relies on idealized assumptions, making complete separation difficult in practice. This can lead to residual spurious correlations that affect generalization performance. In contrast, SIBDNet achieves more thorough feature disentanglement through frequency-domain decomposition based on 3-D DWT and information bottleneck theory, precisely separating spectral–spatial invariant features from domain-sensitive features. This structural disentanglement effectively reduces interference from spurious correlations, thereby significantly enhancing the model's generalization performance.

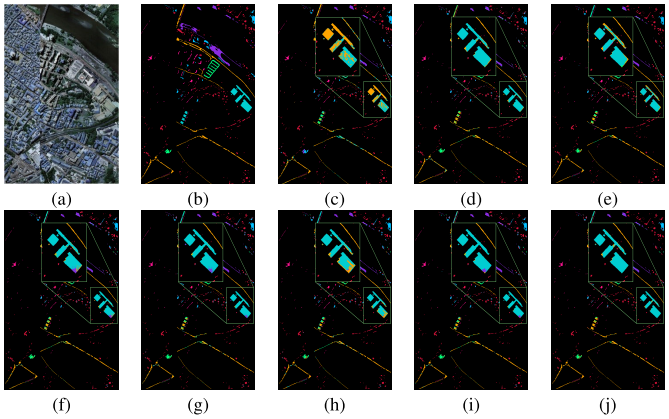


Fig. 2. Classification accuracy maps on PC dataset. (a) False-color image. (b) Ground-truth map. (c) DSNet. (d) SDEnet. (e) LDGnet. (f) LLURnet. (g) FDGNet. (h) ISDGS. (i) ADNet. (j) SIBDNet.

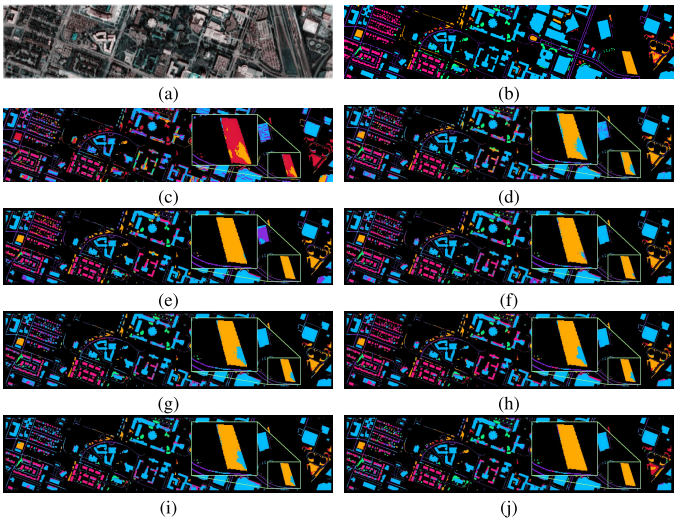


Fig. 3. Classification accuracy maps on Houston18 dataset. (a) False-color image. (b) Ground-truth map. (c) DSNet. (d) SDEnet. (e) LDGnet. (f) LLURnet. (g) FDGNet. (h) ISDGS. (i) ADNet. (j) SIBDNet.

D. Visual Analysis

1) Classification Accuracy Maps: Figs. 2–4 show the classification accuracy graphs of SIBDNet and its comparison methods on the PC dataset, Houston18 dataset, and Loukia dataset. It can be seen that SIBDNet shows the best continuity in the classification accuracy graphs of the three datasets.

2) Misclassification Heat Map: Figs. 5–7 show the misclassification heat maps of SIBDNet and comparative methods on the PC dataset. As evidenced by these results, SIBDNet achieves the lowest misclassification ratio among all

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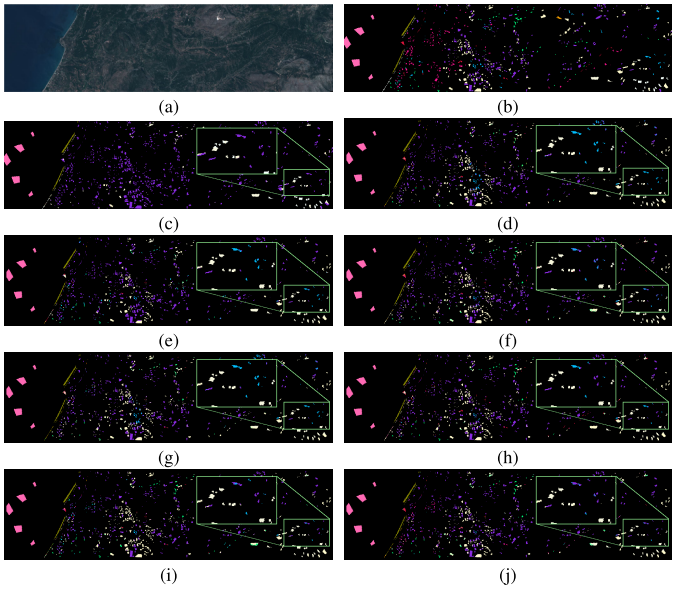


Fig. 4. Classification accuracy maps on Loukia dataset. (a) False-color image. (b) Ground-truth map. (c) DSNet. (d) SDEnet. (e) LDGnet. (f) LLURnet. (g) FDGNet. (h) ISDGS. (i) ADNet. (j) SIBDNet.

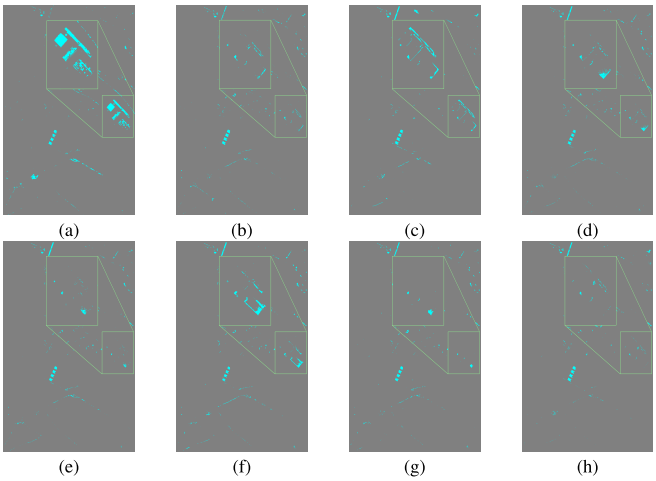


Fig. 5. Misclassification heat map on PC dataset. (a) DSNet. (b) SDEnet. (c) LDGnet. (d) LLURnet. (e) FDGNet. (f) ISDGS. (g) ADNet. (h) SIBDNet.

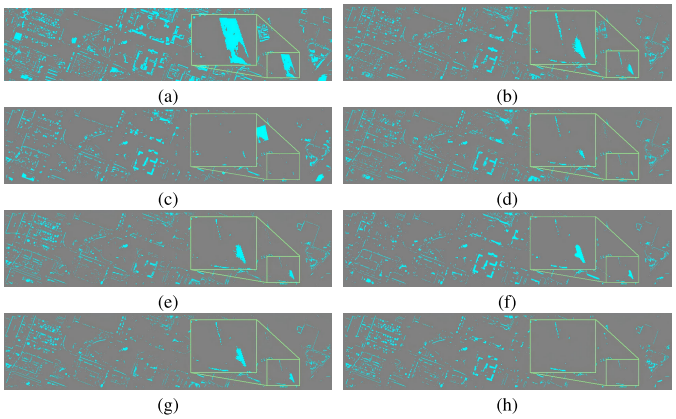


Fig. 6. Misclassification heat map on Houston18 dataset. (a) DSNet. (b) SDEnet. (c) LDGnet. (d) LLURnet. (e) FDGNet. (f) ISDGS. (g) ADNet. (h) SIBDNet.

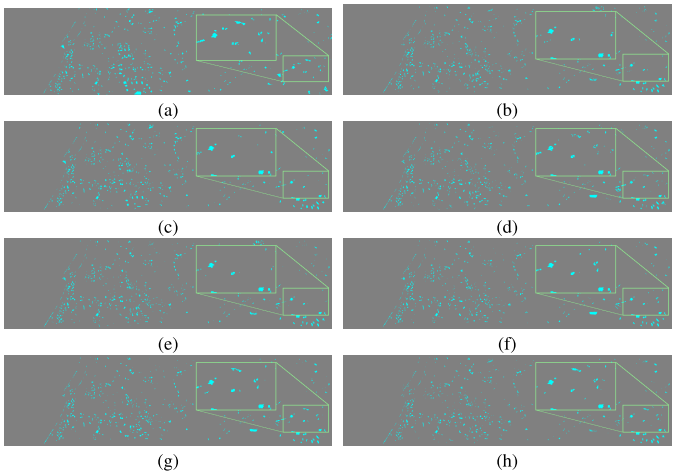


Fig. 7. Misclassification heat map on Loukia dataset. (a) DSNet. (b) SDEnet. (c) LDGnet. (d) LLURnet. (e) FDGNet. (f) ISDGS. (g) ADNet. (h) SIBDNet.

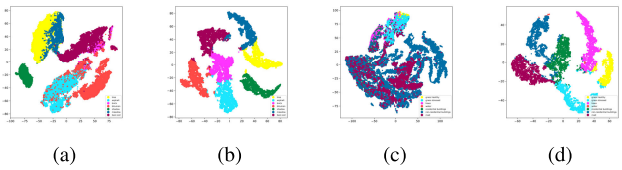
competing methods, further validating its efficacy in cross-scene DG tasks.

From the classification accuracy maps and misclassification heat maps, we can clearly observe that the SIBDNet model demonstrates significant advantages over the comparative methods in several aspects.

1) *Improved Intraclass Consistency:* Within large, homogeneous regions (e.g., Meadows in the PC dataset, Grass stressed in Houston 2018), the classification maps generated by SIBDNet show superior results with notably less noise compared to other methods. This benefit stems from our deep disentanglement mechanism based on 3-D DWT spectral decomposition and the IBE. This mechanism effectively filters out high-frequency noise and spurious correlations induced by DSs, extracting core spectral–spatial features that are truly discriminative for class recognition and invariant across scenes. Consequently, it enables more stable region-level classification on unseen target domains.
2) *Accurate Preservation of Interclass Boundaries and Detailed Structures:* At the junctions between different land-cover classes and for objects with complex internal structures (e.g., residential in the Loukia dataset), the boundaries delineated by SIBDNet are clearer, more aligned with actual object contours, and better preserve their internal spatial details. This is primarily attributed to our Patch-Mix virtual domain generation method. By applying localized, structure-aware perturbations to the disentangled domain-sensitive features, this method allows the model to fully learn a richer set of boundary features and spatial contextual variations during training. This enhances the model's discriminative capability for complex boundaries and local structures, ultimately yielding classification maps with more accurate semantics and finer spatial details.

3) t-distributed stochastic neighbor embedding (t-SNE) Maps: Fig. 8 shows t-SNE visualizations comparing the original HSI with predicted projections of SIBDNet. The

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(a) (b) (c) (d)

Fig. 8. t-SNE map. (a) t-SNE map of original HSI on PC dataset. (b) t-SNE map of projection learned by SIBDNet on PC dataset. (c) t-SNE map of original HSI on the Houston18 dataset. (d) t-SNE map of projection learned by SIBDNet on the Houston18 dataset.

t-SNE map of the original HSI exhibits weak intraclass compactness and interclass separability. In contrast, SIBDNet achieves decoupling of spectral–spatial invariant features and domain-invariant features through spectrum decomposition and information bottleneck disentanglement, thereby mitigating spectral–spatial coupling challenges in HSI. In addition, the Patch-Mix-based virtual domain generation method is employed to transfer disentangled knowledge from the source to the target domain, addressing cross-scene DS. This enhances discriminative feature extraction in target scenes, ultimately improving intraclass consistency and interclass separability of projection features.

E. Ablation Study

SIBDNet integrates three key components: 3-D DWT, IBE, and the Patch-Mix-based virtual domain generation module (Patch-Mix). To validate the contribution of each component, we construct ablation models as detailed in Table VII and conduct comprehensive ablation studies. The design rationale for these ablation variants is elaborated below.

1) *SIBDNet-A:* This model employs conventional separate spatial and spectral feature extraction combined with morphological feature extraction to ensure spectral–spatial feature effectiveness. Spatial and spectral features are randomized using AdaIN to enhance feature diversity, generating virtual domain samples. The model is trained on both source and virtual domains. Once trained, it is directly applied to classify target-domain HSI.
2) *SIBDNet-B:* Based on SIBDNet-A, this model adds the 3-D DWT. It extracts the spectrum decoupling pair $\langle \boldsymbol{H}, \boldsymbol{L} \rangle$ from the frequency domain. Wavelet filters separate these features into LFC and HFC. The HFCs are randomized via AdaIN and then linearly combined with LFC to generate virtual domain samples. The model is subsequently trained on these generated domains for target-scene HSI classification.
3) *SIBDNet-C:* Based on SIBDNet-B, this model adds the IBE. It disentangles LFC and HFC via information bottlenecks, extracting spectral–spatial invariant features and domain-variant features. The domain-variant features undergo randomization through AdaIN before linear recombination with invariant features to generate virtual domain samples. Subsequent training on these generated domains enables direct target-scene HSI classification.
4) *SIBDNet:* Based on SIBDNet-C, this model adds the Patch-Mix. Domain-variant features undergo randomization via Patch-Mix and are then linearly combined with spectral–spatial invariant features to generate virtual domain samples. Subsequent training on these generated domains enables direct zero-shot classification of target-scene HSI.

TABLE VII
ABLATION MODEL

	SIBDNet-A	SIBDNet-B	SIBDNet-C	SIBDNet
3D-DWT	✗	✓	✓	✓
IBE	✗	✗	✓	✓
Patch-Mix	✗	✗	✗	✓

TABLE VIII
IMPACT OF DIFFERENT COMPONENTS

Dataset	Index	SIBDNet-A	SIBDNet-B	SIBDNet-C	SIBDNet
PC	OA(%)	79.94	81.88	83.58	**84.62**
	AA(%)	78.04	80.84	83.19	**85.86**
	Kappa(%)	75.94	78.26	80.31	**81.67**
Houston18	OA(%)	77.87	79.60	80.65	**82.85**
	AA(%)	69.38	72.04	69.16	**78.74**
	Kappa(%)	61.91	64.69	66.88	**71.45**
Loukia	OA(%)	60.07	61.53	63.09	**64.37**
	AA(%)	43.90	45.67	48.95	**51.31**
	Kappa(%)	50.77	52.70	54.53	**56.46**

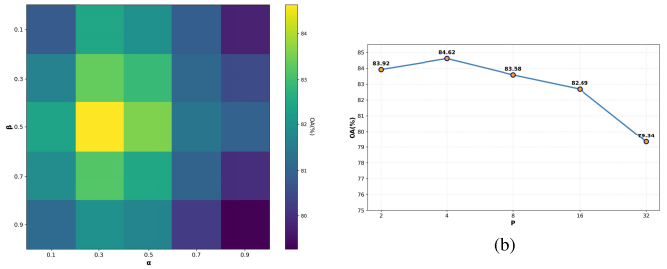
The classification performance of all ablation models across three datasets is detailed in Table VIII. The results reveal that:

Compared to SIBDNet-A, SIBDNet-B achieves classification accuracy improvements of 1.92%, 1.73%, and 1.44% on the three datasets, respectively. This enhancement is attributed to the spectral decomposition via 3-D DWT of SIBDNet-B, which separates high-frequency spectral–spatial coupling information from low-frequency dominant reflectance spectra of ground objects. In contrast, SIBDNet-A lacks mechanisms to decouple class-invariant features from sample-specific domain features, thereby failing to address spectral–spatial coupling challenges during direct feature processing and exacerbating modeling difficulties in high-dimensional HSI data.

Compared to SIBDNet-B, SIBDNet-C achieves classification accuracy improvements of 1.72%, 1.05%, and 1.56% on the three datasets, respectively. This enhancement is attributed to the information bottleneck-based knowledge disentanglement method of SIBDNet-C. This method further disentangles information via spectrum disentanglement pairs composed of LFC and HFC, yielding spectral–spatial invariant features and domain-sensitive features. These features subsequently provide authentic and diverse features essential for cross-scene knowledge transfer.

Compared to SIBDNet-C, SIBDNet achieves classification accuracy improvements of 1.04%, 2.20%, and 1.28% across the three datasets, respectively. This enhancement is attributed to the Patch-Mix-based virtual domain generation method of SIBDNet. This method specifically applies controlled perturbations solely to the domain-sensitive features, thereby expanding the decision boundaries of the feature space. Subsequently, it integrates these perturbed features with the spectral–spatial invariant features to generate virtual domains. This process significantly enhances the effectiveness and reliability of the features, improves the adaptability of

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(a) (b)

Fig. 9. Variation of OA with different parameters. (a) Variation of OA with different α and β. (b) Variation of OA with different P.

cross-domain knowledge transfer to the target domain, and ultimately achieves robust model generalization on the target domain.

F. Parameter Analysis

In this section, we analyzed the impact of different parameter settings on the performance of SIBDNet, including the compression ratio hyperparameter α, the disentanglement degree hyperparameter β, and the patch size P in Patch-Mix. Taking the Pavia dataset as an example, the parameter analysis is as follows.

First, we investigated the influence of α and β on the OA. The color in Fig. 9(a) represents the OA: as the color shifts from deep purple to yellow, the OA value gradually increases. α controls the compression rate of $\boldsymbol{L}$. If α is too large, the compression is too strong, causing $\boldsymbol{L}$ to lose crucial class-discriminative features, which leads to a significant drop in OA. If α is too small, the compression is insufficient, and $\boldsymbol{L}$ will contain a large amount of noise and domain information, resulting in inadequate generalization ability. β controls the disentanglement degree of $\boldsymbol{H}$. If β is too small, the disentanglement is insufficient, and $\boldsymbol{H}$ will retain class information. When perturbing $\boldsymbol{H}$ to generate virtual domains, the class boundaries can be damaged, leading to negative transfer. If β is too large, it forces $\boldsymbol{H}$ to be completely independent of $\boldsymbol{Y}$. An excessively large weight makes optimization difficult and hinders model convergence. The interrelation between α and β and their joint impact on $\langle \boldsymbol{H}, \boldsymbol{L} \rangle$ are as follows

When α is low, and β is high, the model tends to push all information (both class and domain information) into L. This is because doing so simultaneously maximizes $I(\boldsymbol{L}; \boldsymbol{Y})$ and $I(\boldsymbol{X}; \boldsymbol{H}, \boldsymbol{L})$, and $\boldsymbol{L}$ is not penalized by β even if it contains domain information. This renders $\boldsymbol{H}$ ineffective in capturing genuine domain-sensitive features. Consequently, the subsequent Patch-Mix perturbation applied to $\boldsymbol{H}$ becomes ineffective, failing to generate diverse virtual domains and resulting in weak generalization.

When α is high and β is low, $\boldsymbol{L}$ discards many details to meet the compression requirement. To satisfy information completeness and maximize $I(\boldsymbol{X}; \boldsymbol{H}, \boldsymbol{L})$, the information discarded by $\boldsymbol{L}$ must be transferred to $\boldsymbol{H}$. Due to the weak β constraint, some class information leaks into $\boldsymbol{H}$. This causes $\boldsymbol{H}$ to no longer be pure domain features but a mixture with class features. Perturbing $\boldsymbol{H}$ with Patch-Mix then effectively perturbs class features, confusing the classification boundaries and degrading model performance.

TABLE IX
COMPUTATIONAL COST ON EACH DATASET WITH DIFFERENT METHODS

Dataset	Index	DSNet	SDEnet	LDGnet	LLURnet	FDGNet	ISDGS	ADNet	SIBDNet
Pavia	Training Time(s)	5.37	15.57	72.08	22.86	65.28	***10.39***	40.83	135.19
	Test Time(s)	3.98	14.27	22.56	13.59	36.26	10.24	16.07	***4.38***
	FLOPs(G)	0.26	38.18	22.41	***2.30***	11.79	2.39	51.03	33.32
	params(M)	0.25	2.53	36.81	***0.63***	4.23	0.69	2.29	1.45
Houston	Training Time(s)	0.69	6.44	22.47	8.81	13.16	***4.21***	14.18	20.93
	Test Time(s)	4.99	4.65	10.50	6.13	6.54	3.06	9.82	***2.85***
	FLOPs(G)	0.11	35.71	10.49	1.97	3.66	***1.75***	27.26	23.49
	params(M)	0.19	1.88	35.88	0.64	3.29	***0.54***	1.42	1.18
HyRank	Training Time(s)	3.35	10.16	29.39	15.59	9.26	***5.97***	23.76	114.85
	Test Time(s)	1.11	5.58	4.91	4.85	6.65	3.34	4.51	***1.83***
	FLOPs(G)	0.57	41.56	38.46	***3.65***	30.35	3.76	84.48	47.12
	params(M)	0.37	3.42	38.46	***0.68***	5.85	1.50	3.32	1.84

When α is moderate, it forces $\boldsymbol{L}$ to separate out domain information unrelated to $\boldsymbol{Y}$, thereby obtaining domain-invariant class features $\boldsymbol{L}$. When β is moderate, $\boldsymbol{H}$ rejects receiving any information related to $\boldsymbol{Y}$, thus obtaining features $\boldsymbol{H}$ that capture domain variations. This high-quality disentanglement allows the subsequent Patch-Mix to precisely alter only the domain style without changing the class content. The generated virtual domains possess both diversity and semantic consistency, maximizing the DG effect.

Second, we analyzed the parameter P, where P is the patch size in Patch-Mix. Fig. 9(b) shows that as P increases, the OA first increases and then decreases. When P increases from 2 to 4, the OA gradually rises, reaching its maximum at $P = 4$. This is because: an appropriately increased P can encompass the local structures (e.g., edges, textures) typical of ground objects in HSI. The calculated statistics effectively characterize the spectral and spatial properties of the region, allowing for the generation of sufficiently diverse virtual samples while maintaining overall semantic rationality. This better covers potential target domain distributions. However, when P exceeds 8, the OA decreases significantly. This is because an excessively large P value means the patch contains very few pixels. The calculated mean and variance become susceptible to noise and fail to reflect meaningful local spectral–spatial patterns. High-frequency local perturbation essentially adds excessive random noise to the feature space, which may obscure important class-discriminative information and reduce the model's generalization ability.

G. Computational Complexity Analysis

The model complexity of SIBDNet and its comparison methods on three sets of datasets is shown in Table IX, where underlined values indicate the minimum among all methods, and bold-italic values indicate the minimum among the DG methods.

As shown in Table IX, SIBDNet achieves the best performance in terms of test time and classification accuracy, but does not attain optimal performance in training time, FLOPs, and the number of parameters.

Compared to the noncross-domain method DSNet, SIBDNet introduces additional computational overhead to achieve DG capability.

In terms of test efficiency, SIBDNet achieves the best performance among all comparative DG methods, demonstrating the

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high efficiency of our model in practical deployment. However, the training-stage costs (including training time, FLOPs, and parameter count) of our method are not optimal. The specific reasons are as follows.

1) *Cost of Deep Physics-Guided Decomposition:* Unlike methods relying on simple geometric transformations or global style perturbations, our adopted 3-D DWT spectral decomposition requires multiscale filtering and downsampling of high-dimensional HSI cubes in both spatial and spectral dimensions. Although this process introduces computational overhead, its advantage lies in separating low-frequency spectral attributes from high-frequency structural details at the physical level, providing inputs with clear physical meaning for subsequent deep disentanglement. This is difficult to achieve with simple data augmentation methods.
2) *Optimization Complexity of Information Bottleneck-Based Disentanglement:* Our proposed information bottleneck-based knowledge disentanglement architecture is key to achieving high generalization performance. It forces the network to learn compressed class-relevant features and pure domain-sensitive features by optimizing a complex objective function containing mutual information constraints. This process involves variational inference and mutual information boundary estimation, making its optimization more complex than methods that perform direct feature alignment, but it yields superior feature disentanglement results.
3) *Cost of Fine-Grained Virtual Domain Generation:* Our proposed Patch-Mix feature perturbation strategy performs operations such as statistic calculation, shuffling, and perturbation at the image patch level to simulate more realistic interdomain variations. This fine-grained, structure-aware operation works at a finer computational granularity than global style transfer methods (e.g., AdaIN). While it increases overhead, it effectively avoids destroying critical classification boundaries during perturbation, thereby generating higher quality and more diverse virtual domain samples.

In summary, SIBDNet allocates computational complexity to the training stage. Through the three core steps mentioned above, it enhances the model's generalization capability, enabling SIBDNet to achieve the best performance among all comparative DG methods during the test stage. This design philosophy, which emphasizes thorough exploration during training to learn underlying patterns for lightweight and efficient deployment, holds significant practical value for real-world applications with stringent requirements on inference speed, such as on-board remote sensing processing and emergency response.

V. Conclusion

This article proposed the SIBDNet for cross-scene HSI classification. Within SIBDNet, structured high-frequency information and content-representative low-frequency information are initially separated along the spectrum dimension via the 3-D DWT spectrum decomposition method. This process maintains the ability of the model to capture spectral–spatial features while mitigating noise interference caused by coupled spectral–spatial information. Subsequently, the knowledge disentanglement method based on the information bottleneck principle is constructed. This method facilitates the transition from shallow physical disentanglement to deep informational disentanglement, effectively addressing the challenge of coupled spectral–spatial information in HSIs. It thereby provides well-separated spectral–spatial invariant features and domain-sensitive features essential for cross-scene knowledge transfer. Finally, the Patch-Mix-based virtual domain generation method is designed. This method applies controlled perturbations specifically to the domain-sensitive features, thereby broadening the decision boundaries of the feature space. By integrating these perturbed features with the spectral–spatial invariant features to generate virtual domains, it enables effective transfer of the disentangled knowledge from the source domain to the target domain. This process alleviates the DS challenge between source and target domains and enhances the robustness of the model for cross-scene classification. Extensive experiments validate that SIBDNet outperforms current state-of-the-art methods. Ablation studies further demonstrate the effectiveness of its individual components.

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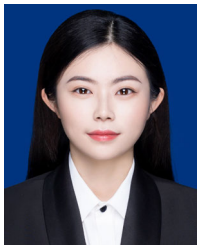
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